

Neuropeptidergic circuit modulation of developmental sleep in *Drosophila*

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The study investigates an emerging research field: the interaction between sleep and development. The authors use *Drosophila* larvae sleep as a study model and provide **valuable** insight into how neuropeptide circuitry controls larvae sleep. By using a broad range of behaviour and imaging methods and analysis, the authors conclude a sleep regulatory neural pathway of Hugin-PK2-Dilps in the *Drosophila* neurosecretory centre IPC. However, the evidence that supports this pathway is **incomplete** - in particular, the methodology in sleep measurement and the specificity at each step of the Hugin-PK2-Dilps pathway require further clarifying experiments or explanation.

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Abstract

Sleep-wakefulness regulation dynamically evolves along development in flies, fish, and humans. While the mechanisms regulating sleep in adults are relatively well understood, little is known about their counterparts in early developmental stages. Here, we report a neuropeptidergic circuitry that modulates sleep in developing *Drosophila* larvae. Through an unbiased screen, we identified the neuropeptide Hugin and its receptor PK2-R1 as critical regulators of larval sleep. Our data suggest that HugPC neurons secrete Hugin peptides to activate insulin-producing cells (IPCs), which express a Hugin receptor PK2-R1. IPCs, in turn, release *Drosophila* insulin-like peptides (Dilps) to regulate sleep. We further show that the Hugin/PK2-R1 axis is dispensable for adult sleep control. Our findings thus reveal the neuromodulatory circuitry regulating larval sleep, highlighting differential impacts of the same modulatory axis on developmental sleep and adult sleep.

Introduction

Sleep is a fundamental state with significant impacts on many aspects of cognition and metabolism (Blumberg et al., 2005 ; Davis et al., 2004 ; Sorribes et al., 2013 ). While sleep remains essential throughout life, its regulation unfolds dynamically over the course of development. For instance, the pace of sleep-wakefulness cycle is relatively fast in infants, but then gradually slows down and stabilizes in adults (Blumberg et al., 2005 ; Davis et al., 2004 .

Sorribes et al., 2013 [↗](#)). Furthermore, sleep-wake cycle in infants is independent of circadian rhythm, but it gradually comes under the circadian regulation (Davis et al., 2004 [↗](#); Frank et al., 2017 [↗](#); Poe et al., 2023 [↗](#)). Such developmental evolution of sleep-wakefulness cycles appears to be conserved across flies and mammals, including humans (Blumberg et al., 2005 [↗](#); Davis et al., 2004 [↗](#); Poe et al., 2023 [↗](#); Szuperak et al., 2018 [↗](#)). In contrast to the deep insight into the molecular and neural mechanisms underlying adult sleep regulation (Barlow & Rihel, 2017 [↗](#); Scammell et al., 2017 [↗](#); Shafer & Keene, 2021 [↗](#)), little is known about regulatory mechanisms underlying infant sleep, partially due to a lack of convenient models to study developmental sleep.

Drosophila larvae have recently emerged as a suitable model for studying sleep regulation mechanisms. Recent studies have reported that the second instar larvae show short periods (< 6 seconds) of an “inactive state” during locomotion. Importantly, this “inactive state” is consistent with the general definition of sleep: reduced responsiveness to noxious stimuli, homeostatic responses to sleep deprivation, and rapid reversibility upon stimulation (Szuperak et al., 2018 [↗](#)), suggesting that the “inactive state” might correspond to the sleep state. Interestingly, loss-of-function mutations in the clock genes *clock* and *cyc* fail to impact larval sleep, while significantly altering the sleep patterns in adult flies (Dubowy & Sehgal, 2017 [↗](#); Hendricks et al., 2003 [↗](#); Szuperak et al., 2018 [↗](#)). Likewise, mutations in the dopamine transporter *DAT* do not influence larval sleep, while reducing sleep amounts in adults (Kume et al., 2005 [↗](#); Szuperak et al., 2018 [↗](#)). These observations imply that sleep regulation mechanisms may be, at least in part, distinct between larvae and adults, yet the neural mechanism of larval sleep remains vastly understudied compared to that of adult sleep.

In this study, we performed an unbiased genetic screen in second-instar larvae, and identified the neuropeptide Hugin and its receptor PK2-R1 as a pair critical for larval sleep control. At the circuit level, Hugin-producing HugPC neurons directly stimulate the insulin-producing cells (IPCs) via PK2-R1 to regulate sleep. Surprisingly, we found that regulators of larval sleep are either irrelevant or exert the opposite effects on adult sleep. Overall, our findings uncover the neuropeptidergic circuitry that regulates larval sleep, and highlight the mechanistic differences between larval and adult sleep.

Results

PK2-R1 is required for sleep control in *Drosophila* larvae

To understand the molecular and neural mechanisms of larval sleep, we modified a previously reported system (Churgin et al., 2019 [↗](#); Szuperak et al., 2018 [↗](#)) to automatically quantify the sleep amount of second instar larvae (**Figure 1—figure supplement 1** [↗](#)) (see Materials & Methods for details). Our system automatically annotates larvae in each frame as either “inactive” or “active” states on the basis of pixel changes, with ~90% accuracy compared to manual annotation (**Figure 1—figure supplement 1C** [↗](#)). We found that bouts of ≥ 12 inactive frames are consistent with the general criteria for sleep: reduced responsiveness to noxious stimuli, homeostatic responses to sleep loss, and rapid reversibility upon stimulation (Raizen et al., 2008 [↗](#); Szuperak et al., 2018 [↗](#)) (**Figure 1—figure supplement 2** [↗](#)). Therefore, the present study defines sleep as an inactive state of ≥ 12 consecutive frames (**Figure 1—figure supplement 1A** [↗](#)).

Using this system, we next carried out an unbiased screen for genes involved in sleep regulation. To this end, we tested 47 CRISPR knock-out lines of genes encoding enzymes or receptors for neuropeptides and monoamines (Deng et al., 2019 [↗](#)). As illustrated in **Figure 1—figure supplement 3** [↗](#), 13 out of 47 homozygous null mutants exhibited significant changes in sleep amounts compared to the control group. To further test the functions of these candidate genes, we next took advantage of the CRISPR knock-in drivers in which GAL4 sequence is inserted into each gene locus. With these drivers, we silenced neurons expressing each candidate gene by expressing

the inward-rectifying potassium channel Kir2.1 (Baines et al., 2001 [↗](#)). Silencing neurons expressing *Oamb* and *PK2-R1* resulted in a significant sleep decrease, phenocopying the knock-out of these genes (Figure 1A [↗](#), 1B [↗](#), and Figure 1—figure supplement 4D [↗](#)). *Oamb* encodes the octopamine receptor and has been reported to be involved in larval sleep (Szuperak et al., 2018 [↗](#)). The role of *PK2-R1* in larval sleep, on the other hand, has been unknown to date, prompting us to focus on this gene.

Insulin-producing cells express PK2-R1 and regulate sleep in developing larvae

Since the *PK2-R1* driver broadly labeled neurons (~1000 cells) throughout the CNS (Figure 1C [↗](#)), we next attempted to identify a subpopulation of these neurons relevant for sleep regulation. Notably, IPCs in the pars intercerebralis (PI), the neurosecretory center of *Drosophila*, have been reported to express *PK2-R1* (Schlegel et al., 2016 [↗](#)). Consistently, the *PK2-R1*^{2A-LexA} knock-in driver labeled all IPCs (Figure 2A [↗](#) and 2B [↗](#)). To assess whether IPCs are involved in larval sleep, we silenced IPCs with Kir2.1 expression using the IPC-specific drivers *Dilp3*- and *Dilp5-GAL4* (Figure 2C [↗](#)). We found that silencing of IPCs significantly increased sleep amounts compared to the control, phenocopying *PK2-R1* neuron silencing and *PK2-R1* knock-out (Figure 1A [↗](#), 1B [↗](#), and 2C [↗](#)). Consistently, both *Dilp3* and *Dilp5* null mutants exhibited larger sleep amounts compared to the control (Figure 2D [↗](#)). These phenotypes of *Dilp* mutants IPC silencing are unlikely to be attributed to locomotion defects, as the travel distances during wake periods were unaffected (Figure 2—figure supplement 1B [↗](#) and 1C [↗](#)). Collectively, these findings suggest that *PK2-R1* in IPCs regulates larval sleep.

Hugin-expressing neurons control larval sleep

Given that PK2-R1 is a receptor for the neuropeptide Hugin (Rosenkilde et al., 2003 [↗](#)), we next examined whether Hugin is involved in larval sleep. We found that *Hug* mutant larvae exhibited significantly larger sleep amounts compared to the control (Figure 3A [↗](#)), consistent with the *PK2-R1* knock-out (Figure 1A [↗](#)). When we silenced *Hug*-expressing neurons by driving the expression of Kir2.1 with the *Hug* CRISPR-knock-in driver *Hug*^{2A-GAL4} (Deng et al., 2019 [↗](#)), it resulted in a significant increase in sleep amounts without detectable changes in locomotion speed (Figure 3B [↗](#) and Figure 3—figure supplement 1B [↗](#)). Conversely, activation of *Hug*-expressing neurons with the red-shifted channel rhodopsin ReaChR (Inagaki et al., 2014 [↗](#)) or the heat-sensitive ion channel TrpA1 (Hamada et al., 2008 [↗](#)) both caused a significant reduction in sleep amount (Figure 3C [↗](#) and 3D [↗](#)). These data together indicate that *Hug*-expressing neurons are responsible for downregulating larval sleep.

Hug-expressing neurons are comprised of 20 cells in the subesophageal zone (SEZ) of the brain, and are classified into four subpopulations based on their projection patterns (Bader et al., 2007 [↗](#)). Of these, one subpopulation called HugPC neurons (Bader et al., 2007 [↗](#)) has been reported to form synaptic connections with IPCs (Hückesfeld et al., 2016 [↗](#)). We thus suspected that HugPC neurons may be responsible for sleep regulation. In line with this notion, axons of HugPC neurons labeled by a newly generated *HugPC-LexA* driver (Figure 3—figure supplement 2 [↗](#)) and IPCs labeled by *Dilp3-GAL4* are located in close proximity to each other (Figure 3E [↗](#)). Indeed, silencing of HugPC neurons by Kir2.1 significantly increased larval sleep (Figure 3F [↗](#)), phenocopying those observed in *Hug* and *PK2-R1* mutants. These results hint at the possibility that HugPC neurons activate IPCs via Hugin/PK2-R1 signaling to regulate sleep.

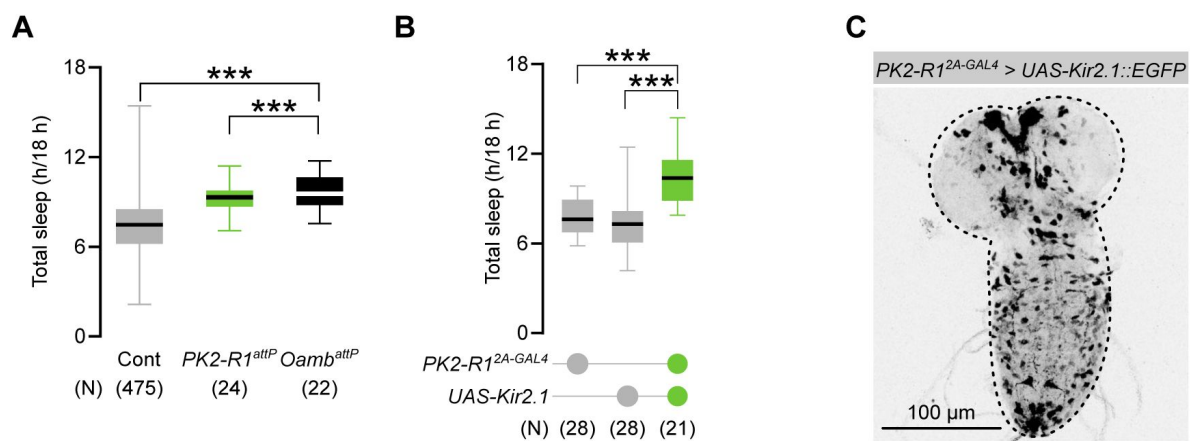


Figure 1.

PK2-R1 is required for larval sleep control

(A) Sleep amounts in *PK2-R1* or *Oamb* knock-out mutants. In this and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.001$ (Mann-Whitney U-test with Bonferroni correction). (B) Sleep amounts in larvae in which *PK2-R1* neurons were silenced. *** $p < 0.001$ (Mann-Whitney U-test with Bonferroni correction). (C) The expression pattern of *PK2-R1^{2A} GAL4 > UAS-Kir2.1::EGFP* in the larval CNS. Similar results were obtained across three independent samples.

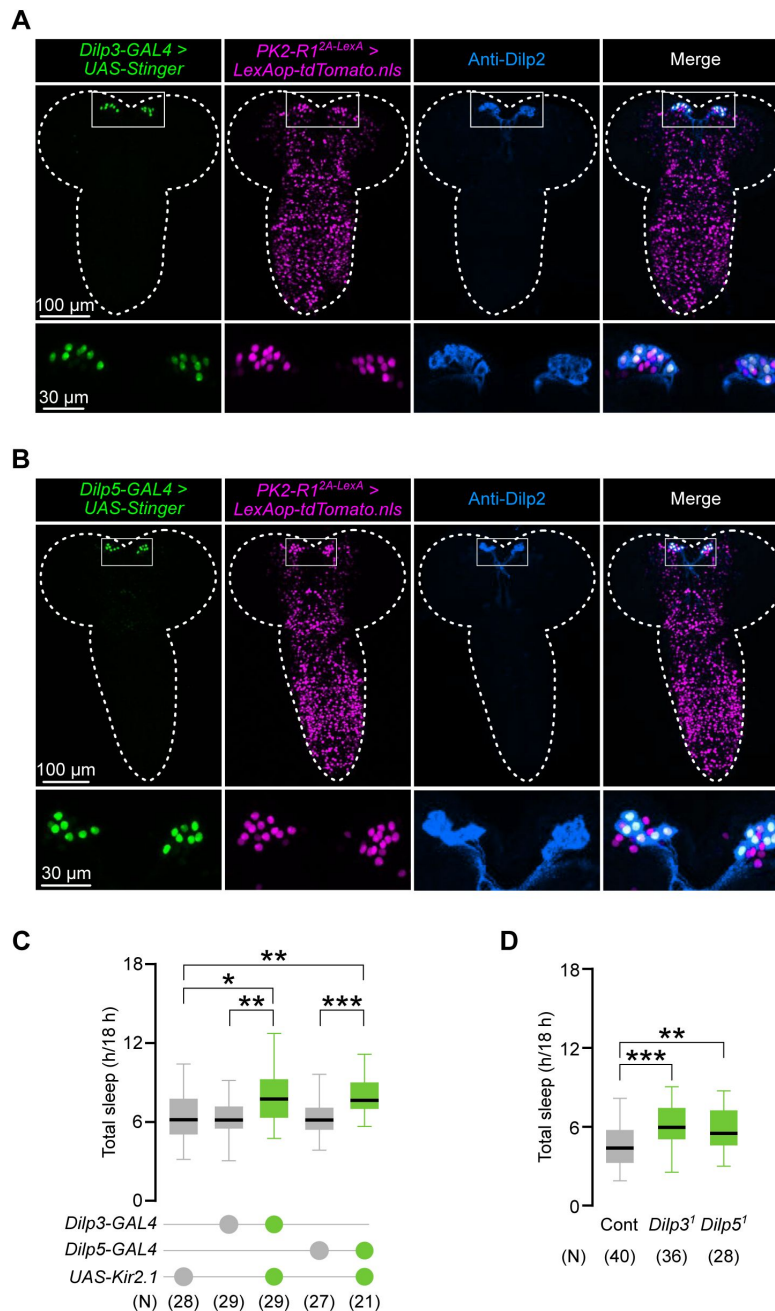


Figure 2.

IPCs and Dilps negatively regulate larval sleep

(A) Triple labeling of *PK2-R1* neurons expressing nuclear-localized RFP (magenta), *Dilp3* neurons expressing nuclear-localized GFP (green), and anti-Dilp2 positive cells (blue). Top panels show signals in the larval brain and the VNC. The bottom panels show magnified images of the white-squared area in the top panels, where the cell bodies of the IPCs are located. Note that all IPCs labeled by *Dilp3-GAL4* are also labeled by *PK2-R1^{2A-LexA}*. Similar results were obtained across five independent samples. (B) Triple labeling of *PK2-R1* neurons expressing nuclear-localized RFP (magenta), *Dilp5* neurons expressing nuclear-localized GFP (green), and anti-Dilp2 positive cells (blue). Similar results were obtained across five independent samples. (C) Effect of IPCs silencing on larval sleep. In this and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (Mann-Whitney U-test with Bonferroni correction). (D) Sleep amounts in *Dilp3* or *Dilp5* null mutant larvae. *** $p < 0.001$, ** $p < 0.01$ (Mann-Whitney U-test with Bonferroni correction).

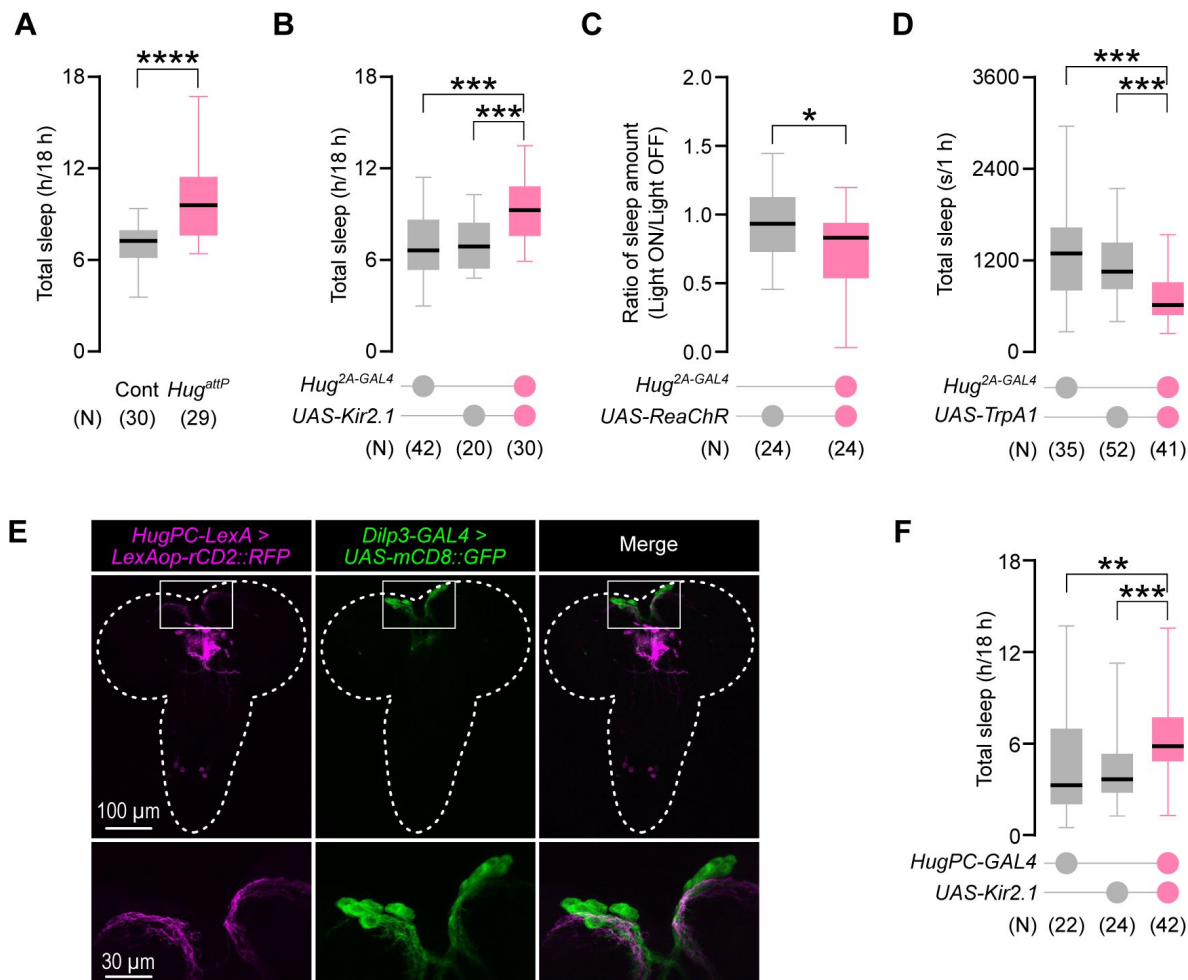


Figure 3.

Hug neurons negatively regulate larval sleep

(A) Sleep amounts in *Hug* CRISPR-knock-out mutant larvae. In this and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. **** p < 0.0001 (Mann-Whitney U-test). (B) Effect of silencing *Hug* neurons on larval sleep amount. *** p < 0.001 (Mann-Whitney U-test with Bonferroni correction). (C) Larval sleep changes induced by optogenetic activation of *Hug* neurons. For each genotype, sleep duration in the 1-h light-ON period was normalized to that in the 1-h light-OFF phase. * p < 0.05 (Mann-Whitney U-test). (D) Larval sleep during thermogenetic activation of *Hug* neurons. *** p < 0.001 (Mann-Whitney U-test with Bonferroni correction). (E) Images of HugPC neurons and IPCs labeled by rCD2::RFP (magenta) and mCD8::GFP (green), respectively. The bottom panels show magnified images of the white-squared region in the top panels, in which HugPC neurons project their axons close to the cell bodies of the IPCs. While the top panels are z-stacks of 122 image slices (1 μ m interval) covering the entire brain, the bottom panels are projections of 60 slices around the cell bodies of IPCs. Similar results were obtained across five independent samples. (F) Effect of silencing HugPC neurons on larval sleep. *** p < 0.001, ** p < 0.01 (Mann-Whitney U-test with Bonferroni correction).

Hugin triggers Ca^{2+} elevation and Dilp secretion in larval IPCs via PK2-R1

A previous report showed that IPCs exhibit increased firing rates and intracellular calcium (Ca^{2+}) levels upon stimulation (Kr neisz et al., 2010). This motivated us to test whether Hugin activates IPCs. To this end, we utilized the recently developed calcium indicator CRTC::GFP, which can report Ca^{2+} dynamics in the timescale of several minutes (Bonheur et al., 2023). As glucose reportedly triggers Ca^{2+} dynamics in IPCs in a range of minutes (Oh et al., 2019), we wondered if CRTC::GFP can detect these dynamics as a positive control. To this end, we applied D-glucose to isolated larval brains and found a significant cytosol-to-nuclear translocation of CRTC::GFP in IPCs within ~5 min, confirming that this reporter can indeed detect Ca^{2+} responses in IPCs upon stimulation (Figure 5A and 5B). We then examined whether *Hug* neurons act upstream of PK2-R1-expressing IPCs, by measuring Ca^{2+} responses of IPCs while thermogenetically activating *Hug* neurons with TrpA1 (Figure 4A). In both isolated brains (*ex vivo*) and living larvae (*in vivo*), we found that activation of *Hug* neurons significantly increased Ca^{2+} levels in IPCs (Figure 4B and 4C), suggesting that *Hug* neurons act upstream of IPCs (Figure 4D). Next, we tested whether Hug peptides can induce Ca^{2+} responses in IPCs using *ex vivo* preparations. The *Hug* gene encodes a precursor that potentially produces at least two distinct peptides, Hug-  and PK-2, both of which can bind PK2-R1 to induce Ca^{2+} responses (Rosenkilde et al., 2003). We thus incubated the CRTC::GFP-expressing larval brain with chemically synthesized Hug-  or PK-2 (Figure 5A). We found that bath application of either Hug-  or PK-2 induced significant Ca^{2+} responses in IPCs (Figure 5B–D). In agreement with the idea that Hug peptides activate IPCs via PK2-R1, knock-out of PK2-R1 blocked the Hug peptide-evoked Ca^{2+} elevation in IPCs (Figure 5G). Similar results were obtained when we bath-applied the synthetic Neuromedin U (NMU), the mammalian homologue of Hugin (Figure 5A, 5C, and 5G), suggesting that the role of Hugin/PK2-R1 signaling may be conserved across species. Furthermore, Dilp3 accumulated in IPCs was significantly reduced after applying Hug peptides, suggesting that Hugin causes IPCs to release Dilp3 (Figure 5E). Collectively, these data suggest that *Hug* neurons activate IPCs via Hugin/PK2-R1 signaling to regulate sleep (Figure 5H).

Hugin/PK2-R1 axis has distinct impacts on sleep control in larvae and adults

Given the differences between larval versus adult sleep in temporal pattern and circadian influence, we next wondered if the Hugin/PK2-R1/Dilps axis functions in adult sleep as well. As in larvae, PK2-R1 knock-out adults showed increased sleep amount compared to the control (Figure 6—figure supplement 1B). In contrast, *Hug* knock-out or silencing of HugPC neurons failed to influence adult sleep (Figure 6A and 6B). This suggests that, unlike in larvae, Hug is dispensable for adult sleep.

Interestingly, the expression patterns of PK2-R1 and *Hug*, as well as the morphology of HugPC neurons and IPCs, were almost comparable between larvae and adults (Figure 6—figure supplement 2). This implies that the differential roles of Hug in larvae versus adults are likely due to physiological differences in HugPC neurons and/or IPCs. We thus examined how Hug peptides evoke Ca^{2+} responses in adult IPCs and found that neither Hug-  nor PK-2 could induce Ca^{2+} responses in adult IPCs (Figure 6D and 6E), unlike in larvae. Consistently, Hug peptides failed to reduce Dilp3 levels in adult IPCs (Figure 6F). These data indicate that Hug induces Ca^{2+} responses and Dilp secretion in larval IPCs, but not in adult IPCs. Surprisingly, we further found that *Dilp3* or *Dilp5* null mutations reduced sleep amounts in adults (Figure 6C), while in larvae, these mutations in fact increased the sleep amounts. These data unexpectedly reveal how the same set of neuronal circuits and molecular signaling therein can evolve over development in a complex manner.

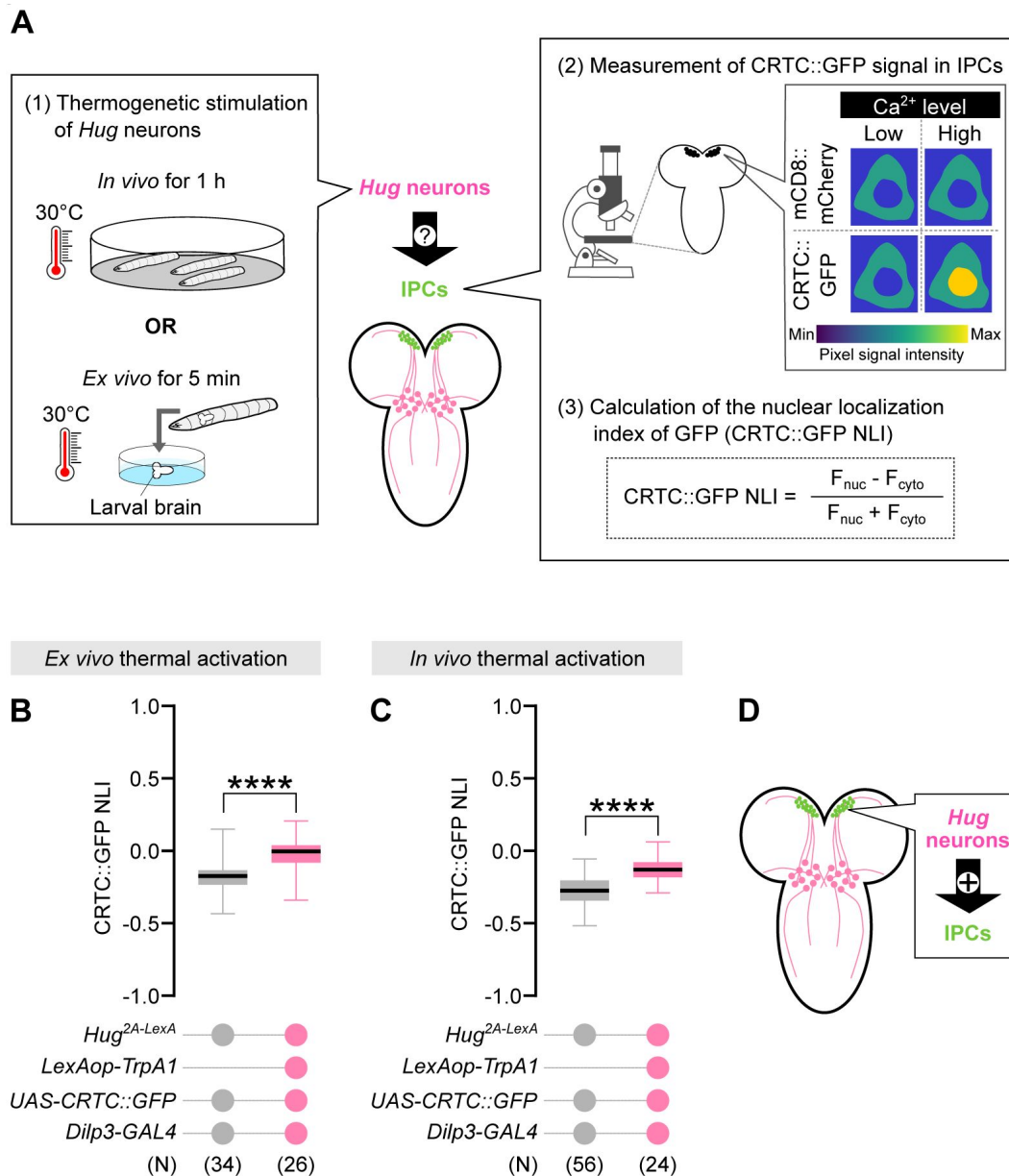


Figure 4.

Activation of *Hug* neurons triggers Ca²⁺ responses in larval IPCs

(A) Schematic flow of assessing intracellular Ca²⁺ levels in IPCs while performing the thermogenetic activation of *Hug* neurons. (B–C) The level of nuclear-localized CRTC::GFP signal, calculated as NLI (see the Methods section for a detailed explanation of NLI), in IPCs during *Hug* neurons stimulation *ex vivo* (B) or *in vivo* (C). In these panels, 'N' indicates the number of cells used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. **** $p < 0.0001$ (Mann-Whitney U-test). (D) The proposed neuronal connection that regulates larval sleep.

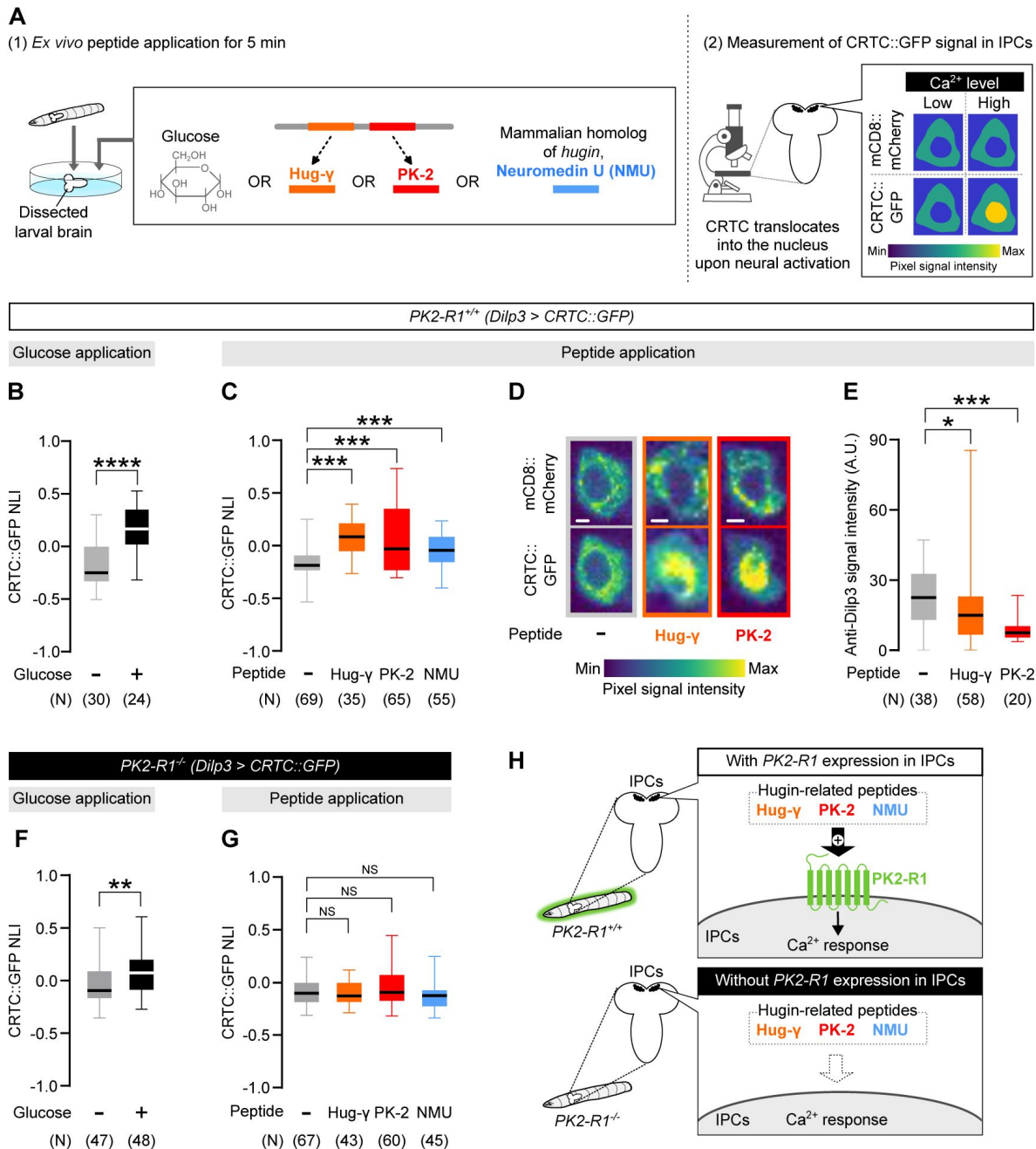


Figure 5.

Hug peptides induce Ca²⁺ responses in larval IPCs via PK2-R1

(A) Schematic flow of assessing intracellular Ca²⁺ levels in IPCs and bath application of peptides. (B–C) Ca²⁺ responses in IPCs to *ex vivo* bath application of either glucose (B) or Hug peptides (C). In this and the following panels, ‘N’ indicates the number of cells used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. **** $p < 0.0001$, *** $p < 0.001$ (Mann–Whitney U-test with Bonferroni correction). (D) Representative images of larval IPCs after peptide application. Scale bars, 2 μ m. (E) Anti-Dilp3 signal intensity measured within the cytosolic areas of IPCs. *** $p < 0.001$, * $p < 0.05$ (Mann–Whitney U-test with Bonferroni correction). (F–G) Ca²⁺ responses in IPCs of PK2-R1 knock-out mutants, measured following either glucose (F) or Hug peptide application (G). ** $p < 0.01$, NS: $p \geq 0.05$ (Mann–Whitney U-test with Bonferroni correction). (H) A model where Hug peptides, but not glucose, activate IPCs via the PK2-R1 receptor.

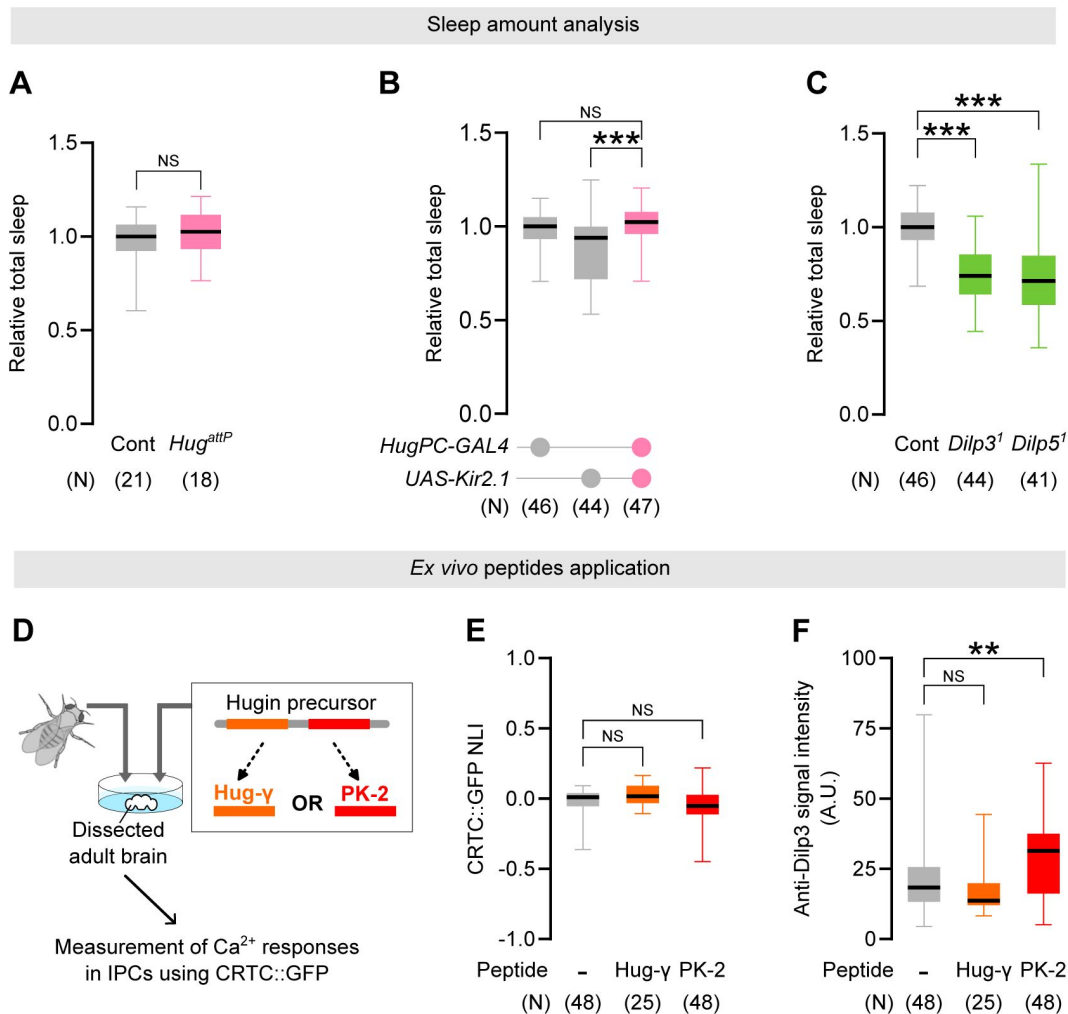


Figure 6.

Distinct impacts of Hugin/PK2-R1 axis on wake/sleep control in larvae and adults

(A) Total sleep amounts in *Hug* knock-out mutant adults. NS: $p \geq 0.05$ (Mann-Whitney U-test). In these and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. NS: $p \geq 0.05$ (Mann-Whitney U-test). (B) Sleep amounts of adults in which *HugPC* neurons were silenced. *** $p < 0.001$, NS: $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction). (C) Sleep amounts in *Dilp3* or *Dilp5* null mutant adults. *** $p < 0.001$ (Mann-Whitney U-test with Bonferroni correction). (D) Schematic flow of peptide application experiments using adult brains. (E-F) Ca^{2+} responses (E) and anti-Dilp3 signal intensity (F) in adult IPCs after Hug peptide application. ** $p < 0.01$, NS: $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction).

Discussion

In this study, we have identified the neuropeptide Hugin and its receptor PK2-R1 as a ligand/receptor pair critical for sleep regulation in *Drosophila* larvae. We have further shown that larval IPCs express PK2-R1, respond to and release Dilp3 upon Hugin stimulation. Surprisingly, the Hugin/PK2-R1 axis is dispensable for sleep regulation in adults, even though gene expressions as well as the circuit structure appeared to be conserved between larval and adult brains. Furthermore, Dilps appear to modulate sleep in opposite directions in larvae and adults. Our findings thus uncover the neuropeptidergic modulation circuitry that specifically regulates developmental sleep, and suggest divergent usage of the same molecules and circuitry for sleep modulation in larvae and adults.

Hugin/PK2-R1/Dilps axis negatively regulates sleep in larvae

The present study shows that the neuropeptide Hugin and its receptor PK2-R1 are required for sleep regulation in the early larval stage. This notion is supported by the following lines of evidence. First, knock-out mutations in *Hug* or *PK2-R1* increased sleep amounts compared to the control (**Figure 1A** and **3A**). Consistently, silencing a subpopulation of *Hug*-expressing neurons, called HugPC neurons, or *PK2-R1* neurons increased sleep amounts (**Figure 1B** and **3F**). Second, thermogenetic activation of *Hug* neurons decreased sleep amounts (**Figure 3D**). Third, HugPC neurons project their axons to IPCs in the brain that express PK2-R1 receptors (**Figure 3E**). Fourth, Hug peptides induced Ca^{2+} elevation and promoted Dilp3 release from IPCs in a PK2-R1- dependent manner (**Figure 5**). Last, *Dilp3* null mutations increased sleep amounts (**Figure 2D**). Based on these data, we propose a model in which the Hugin/PK2-R1/Dilps axis along the HugPC-IPC circuitry modulates larval sleep (**Figure 7**).

Examples of Hugin acting through PK2-R1 are prominent in the context of feeding regulation in larvae and adults (Schoofs et al., 2014). For example, a recent study has suggested that Hugin regulates the timing of pupariation and body wall contraction during pupariation through PK2-R1 in PTTH neurons (Ohhara et al., 2024). In the present study, we showed that HugPC in the larval brain modulates sleep (**Figure 3F**). A previous electron microscopic analysis suggested that larval HugPC neurons likely form synaptic connections to downstream neurons, including IPCs (Hückesfeld et al., 2021; Schlegel et al., 2016). We here showed that larval sleep control requires PK2-R1, and that thermogenetic activation of *Hug* neurons causes Ca^{2+} responses in PK2-R1- expressing IPCs (**Figure 4**). Importantly, our *ex vivo* data indicated that Hugin peptide-induced Ca^{2+} response in IPCs was largely abolished by *PK2-R1* mutation (**Figure 5**), indicating the critical role of the Hugin/PK2-R1 axis for the HugPC-mediated IPC activation. Together, we propose that the HugPC-IPCs circuitry mobilizes Hugin/PK2-R1 signaling to selectively modulate larval sleep. Interestingly, our *ex vivo* data indicated that Neuromedin U (NMU), a mammalian ortholog of Hugin, can trigger Ca^{2+} responses in larval IPCs in a PK2-R1-dependent fashion (**Figure 5**). Given that NMU has been implicated in sleep regulation in fish and mammals (Ahnaou & Drinkenburg, 2011; Chiu et al., 2016), an analogous ligand/receptor pair might be involved in sleep modulation in mammals as well.

Divergent roles of insulin in regulation of larval sleep and feeding

IPCs in larvae responded to exogenously applied Hug peptides, both by exhibiting elevated Ca^{2+} levels and decreases in intracellular Dilp3 storage (**Figure 5C–E**). In contrast to Dilp3, however, the amount of Dilp2 accumulated in IPCs did not decrease following Hug peptide application (**Figure 5—figure supplement 1B**). On the other hand, Dilp2 accumulation in IPCs decreased

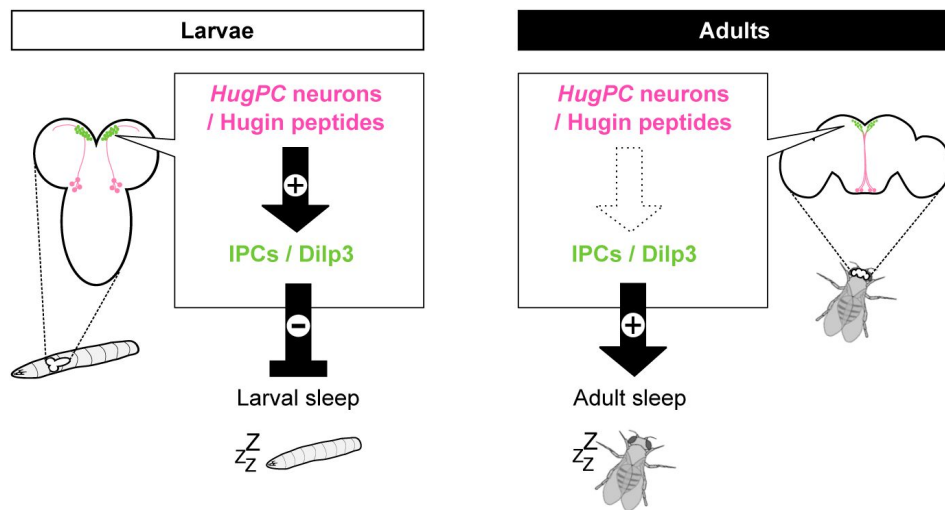


Figure 7.

A schematic model of the Hugin/PK2-R1/Dilps axis in larval and adult sleep regulation

after glucose treatment (**Figure 5—figure supplement 1A**) (Oh et al., 2019). These results unexpectedly suggest that larval IPCs form parallel channels inside, one for Hug-Dilp3 and the other for glucose-Dilp2.

Mechanistically, Hug acts directly on IPCs via the PK2-R1 receptor. As for glucose, on the other hand, glucose-sensing CN neurons in the brain take up glucose and subsequently secrete another neuropeptide to activate IPCs (Oh et al., 2019). Such parallelism inside IPCs might provide insights into why insect species broadly express multiple insulin-like peptide genes, in some cases even as many as 40 genes in the silkworm *Bombyx mori* (Okamoto, 2021). *Drosophila* genome harbors eight *Dilp* genes (Nässel et al., 2015), of which *Dilp2*, *Dilp3*, and *Dilp5* are considered nutrient-related.

Nevertheless, mRNA expression or release patterns of these Dilps are not identical under different nutrient conditions (Géminard et al., 2024; Nässel et al., 2015; Oh et al., 2019; Ikeya et al., 2002). Furthermore, these Dilps exhibit different binding affinities for the insulin receptor (InR), the secreted decoy of InR (SDR), and the ecdysone-inducible gene L2 (Imp-L2) (Okamoto et al., 2013). These reports hint at the differential roles of different insulin-like peptides; however, mechanistic insights remain largely lacking. Our data on larval IPCs forming parallel channels for Hug-Dilp3 and glucose-Dilp2 may provide a point of entry for tackling this mystery.

Differential impacts of the same neuropeptidergic modulations on larval and adult sleep

Our data show that the Hugin/PK2-R1/Dilps axis likely downregulates sleep amounts in larvae. Despite the significant increase in sleep amount in *Hug* mutant larvae, *Hug* mutant adults showed sleep amounts comparable to wild-type control (**Figure 6A**), consistent with a previous report that Hugin is dispensable for control of baseline sleep levels (Schwarz et al., 2021). According to our histological analyses, the Hugin/PK2-R1 expression patterns, as well as the HugPC-IPCs circuitry, appear to be largely conserved in adults (**Figure 3E** and **Figure 6—figure supplement 2**). Unlike larval brains, however, Hug peptides failed to trigger Ca^{2+} responses in adults IPCs (**Figure 6E**). It is thus possible that Hugin/PK2-R1 signaling along the HugPC-IPCs circuitry is suppressed in adults. IPCs in adults receive multiple positive and negative modulatory inputs through GPCRs including the metabotropic GABA_B receptors (Enell et al., 2010), which suppress IPC activity and Dilp release in adult IPCs (Enell et al., 2010). It is thus plausible that such negative modulatory inputs to IPCs in adults might counteract the Hugin/PK2-R1 axis to suppress Dilp release. In addition, our data suggest that Dilps modulate sleep amount in the opposite directions in larvae and adults (**Figure 7**).

Comparing the expression levels and activities of GPCRs in larval and adult IPCs would be essential to better understand how same modulatory signals over the course of development come to exert differential impacts on sleep. Interestingly, Hugin in adults appears irrelevant for the baseline sleep amount but is required for homeostatic regulation of sleep (Schwarz et al., 2021). Thus, testing whether the Hugin/PK2-R1 axis is involved in the homeostatic regulation of larval sleep, and how this system compares to its adult counterpart, may further provide mechanistic insights into how homeostatic sleep regulation matures over development.

In summary, we identified the Hugin/PK2-R1/Dilps signaling along the HugPC/IPCs circuit as the molecular and circuitry basis for larval sleep regulation. Our study begins to fill the knowledge gap in larval sleep regulation, and sheds light on mechanistic differences in sleep regulation between early developmental stages and adults.

Materials and methods

Key resource table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Gene (<i>Drosophila melanogaster</i>)	<i>PK2-R1</i>	NA	FLYB:FBgn0038140	
Gene (<i>D. melanogaster</i>)	<i>Dilp3</i>	NA	FLYB:FBgn0044050	
Gene (<i>D. melanogaster</i>)	<i>Dilp5</i>	NA	FLYB:FBgn0044048	
Gene (<i>D. melanogaster</i>)	<i>Hug</i>	NA	FLYB:FBgn0028374	
Gene (<i>Homo sapiens</i>)	<i>NMU</i>	NA	HGNC:7859	
Genetic reagent (<i>D. melanogaster</i>)	<i>w¹¹¹⁸</i>	Other	NA	Provided by Dr. Yi Rao, Peking University.
Genetic reagent (<i>D. melanogaster</i>)	<i>iso31</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:5905; RRID:BDSC_5905	Isogenized for chr 1;2;3, with w[1118] line.
Genetic reagent (<i>D. melanogaster</i>)	<i>PK2-R1^{attP}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84563; RRID:BDSC_84563	FlyBase symbol: TI{TI}PK2-R1[attP]
Genetic reagent (<i>D. melanogaster</i>)	<i>PK2-R1^{2A-GAL4}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84686; RRID:BDSC_84686	FlyBase symbol: TI{2A-GAL4}PK2-R1[2A-GAL4]
Genetic reagent (<i>D. melanogaster</i>)	<i>PK2-R1^{2A-LexA}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84431; RRID:BDSC_84431	FlyBase symbol: TI{2A-lexA::GAD}PK2-R1[2A-lexA]
Genetic reagent (<i>D. melanogaster</i>)	<i>Dilp3^l</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:30882; RRID:BDSC_30882	FlyBase symbol: TI{TI}Ilp3[1]
Genetic reagent (<i>D. melanogaster</i>)	<i>Dilp5^l</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:30884; RRID:BDSC_30884	FlyBase symbol: TI{TI}Ilp5[1]
Genetic reagent (<i>D. melanogaster</i>)	<i>Dilp3-GAL4</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:52660; RRID:BDSC_52660	FlyBase symbol: P{Ilp3-GAL4.C}2

Genetic reagent (<i>D. melanogaster</i>)	<i>Dilp5-GAL4</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:66007; RRID:BDSC_66007	FlyBase symbol: P{Ilp5-GAL4.L}8
Genetic reagent (<i>D. melanogaster</i>)	<i>Hug^{attP}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84514; RRID:BDSC_84514	FlyBase symbol: TI{TI}Hug[attP]
Genetic reagent (<i>D. melanogaster</i>)	<i>Hug^{2A-GAL4}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84646; RRID:BDSC_84646	FlyBase symbol: TI{2A-GAL4}Hug[2A-GAL4]
Genetic reagent (<i>D. melanogaster</i>)	<i>Hug^{2A-LexA}</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:84397; RRID:BDSC_84397	FlyBase symbol: TI{2A-LexA::GAD}Hug[2A-LexA]
Genetic reagent (<i>D. melanogaster</i>)	<i>HugPC-GAL4</i>	Hückesfeld et al., 2016 (DOI:10.1038/ncomms12796)	NA	Provided by Dr. Michael J. Pankratz, University of Bonn.
Genetic reagent (<i>D. melanogaster</i>)	<i>HugPC-LexA</i>	This paper	NA	The 544-bp enhancer region of Hug was placed upstream of nlsLexA::p65, and the construct was inserted into attP2 site by ΦC31 integrase-mediated transgenesis. See Methods for details.
Genetic reagent (<i>D. melanogaster</i>)	<i>UAS-Kir2.1::EGFP</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:6596; RRID:BDSC_6596	FlyBase symbol: P{UAS-

				HsapYKCNJ2.E GFP}1
Genetic reagent (<i>D. melanogaster</i>)	<i>UAS-Stinger</i> , <i>LexAop-tdTomato.nls</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:66680; RRID:BDSC_66680	FlyBase symbol: P{UAS-Stinger}2; PBac{13XLexAop2-IVS-tdTomato.nls}VK00022
Genetic reagent (<i>D. melanogaster</i>)	<i>UAS-mCD8::mCherry</i> , <i>UAS-CRTC::GFP</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:99657; RRID:BDSC_99657	FlyBase symbol: P{UAS-mCD8.mCherry-T2A-lacZ.nls}JK22C; P{UAS-Crtc.GFP}attP40
Genetic reagent (<i>D. melanogaster</i>)	<i>UAS-ReaChR</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:53741; RRID:BDSC_53741	FlyBase symbol: P{UAS-ReaChR}attP40
Genetic reagent (<i>D. melanogaster</i>)	<i>UAS-TrpA1</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:26263; RRID:BDSC_26263	FlyBase symbol: P{UAS-TrpA1(B).K}attP16
Genetic reagent (<i>D. melanogaster</i>)	<i>LexAop-TrpA1</i>	Burke et al., 2012 (DOI:10.1038/nature11614)	NA	Provided by Dr. Scott Waddell, University of Oxford.
Genetic reagent (<i>D. melanogaster</i>)	<i>LexAop-rCD2::RFP-p10.UAS-mCD8::GFP-p10</i>	Bloomington <i>Drosophila</i> Stock Center	BDSC:67093; RRID:BDSC_67093	FlyBase symbol: P{lexAop-rCD2::RFP-p10.UAS-mCD8::GFP-p10}su(Hw)attP5

Genetic reagent (<i>Escherichia coli</i>)	One Shot™ ccdB Survival™ 2 T1 ^R competent cells	Thermo Fisher Scientific	Thermo Fisher Scientific:A104 60	
Antibody	Anti-GFP (chicken polyclonal)	Aves Labs	Aves Labs:GFP- 1020; RRID:AB_1000 0240	IHC (1:1000)
Antibody	Anti-mCherry (mouse monoclonal)	Takara Bio	Takara Bio:632543; RRID:AB_2307 319	IHC (1:1000)
Antibody	Anti-Dilp3 (rabbit polyclonal)	Veenstra et al., 2009 (DOI:10.1007/s0 0441-009-0769- y)	NA	IHC (1:250) Provided by Dr. Jan A. Veenstra, Université de Bordeaux.
Antibody	Anti-Dilp2 (rabbit polyclonal)	Okamoto et al., 2012 (DOI:10.1073/p nas.1116050109)	NA	IHC (1:2000) Provided by Dr. Takashi Nishimura, RIKEN.
Antibody	Anti-brp (mouse monoclonal)	Developmental Studies Hybridoma Bank	DSHB:nc82; RRID: AB_2314866	IHC (1:100)
Antibody	Anti-histone H3 phospho S28 (rat monoclonal)	Abcam	Abcam:ab10543 ; RRID:AB_2295 065	IHC (1:1000)
Antibody	Goat anti- chicken Alexa Fluor 488	Thermo Fisher Scientific	Thermo Fisher Scientific:A- 11039; RRID:AB_2534 096	IHC (1:200)

Antibody	Goat anti-mouse Alexa Fluor 555	Thermo Fisher Scientific	Thermo Fisher Scientific:A- 21424; RRID:AB_1417 80	IHC (1:200)
Antibody	Goat anti-rabbit Alexa Fluor 633	Thermo Fisher Scientific	Thermo Fisher Scientific:A- 21071; RRID:AB_2535 732	IHC (1:200)
Antibody	Goat anti-rat Alexa Fluor 633	Thermo Fisher Scientific	Thermo Fisher Scientific:A- 21094; RRID:AB_2535 749	IHC (1:200)
Recombinant DNA reagent	pBPnlsLexA::p6 5Uw	Addgene	Addgene:26230; RRID:Addgene_ 26230	
Sequence-based reagent	<i>HugPC</i> enhancer	This paper	NA	A 544-bp fragment PCR- amplified from the fly genomic DNA. See Methods for details.
Peptide, recombinant protein	Hug- γ	This paper	NA	Chemically synthesized <i>Drosophila</i> Hug- γ peptide with amino acid sequences of Acetyl- QLQSNGEPAY RVRTPL- CONH ₂

Peptide, recombinant protein	PK-2	This paper	NA	Chemically synthesized <i>Drosophila</i> PK- 2 peptide with amino acid sequences of Acetyl- SVPFKPRL- CONH2
Peptide, recombinant protein	NMU	This paper	NA	Chemically synthesized human NMU peptide with amino acid sequences of Acetyl- FRVDEEFQSP FASQSRGYFL FRPRN- CONH2
Commercial assay or kit	PrimeSTAR Max DNA Polymerase	Takara Bio	Takara Bio:R045A	
Commercial assay or kit	In-Fusion HD Cloning Kit w/Cloning Enhancer	Takara Bio	Takara Bio:639635	
Chemical compound, drug	D(+)-Glucose	FUJIFILM Wako Pure Chemical Corporation	FUJIFILM Wako Pure Chemical Corporation:041 -00595	
Software, algorithm	Adobe Illustrator 2023	Adobe	RRID:SCR_010 279	
Software, algorithm	Adobe Photosop 2023	Adobe	RRID:SCR_014 199	

Software, algorithm	ImageJ v1.53t	Schneider et al., 2012 (DOI:10.1038/nmeth.2089)	RRID:SCR_003070	
Software, algorithm	Inkscape v1.2	The Inkscape Team	RRID:SCR_014479	
Software, algorithm	Prism v9.5.1	GraphPad	RRID:SCR_002798	
Software, algorithm	Python Programming Language v3.10.0	Python Software Foundation	RRID:SCR_008394	
Software, algorithm	R Project for Statistical Computing v4.1.2	R Core Team 2021	RRID:SCR_001905	
Software, algorithm	SnapGene v5.3.3	GSL Biotech LLC	RRID:SCR_015052	
Software, algorithm	Arduino UNO	Arduino	RRID:SCR_017284	
Software, algorithm	TriKinetics DAMSystem3 Software	TriKinetics Inc. (available at https://trikinetics.com/)	RRID:SCR_021809	
Other	TriKinetics Drosophila Activity Monitoring System	TriKinetics Inc.	RRID:SCR_021798	
Other	Fan-less Peltier-type incubator	Mitsubishi Electric Engineering Co., Ltd.	Mitsubishi Electric Engineering Co., Ltd.:SLC-25	Used to keep flies for DAM experiments.

Fly strains

The following strains of *Drosophila melanogaster* were obtained from the Bloomington *Drosophila* Stock Center: *iso31* (an isogenic *w*¹¹¹⁸ strain; BDSC# 5905), *PK2-R1^{attP}* (BDSC# 84563), *PK2-R1^{2A-GAL4}* (BDSC# 84686), *PK2-R1^{2A-LexA}* (BDSC# 84431), *Dilp3¹* (BDSC# 30882), *Dilp5¹* (BDSC# 30884), *Dilp3-GAL4* (BDSC# 52660), *Dilp5-GAL4* (BDSC# 66007), *Hug^{attP}* (BDSC# 84514), *Hug^{2A-GAL4}* (BDSC# 84646), *Hug^{2A-LexA}* (BDSC# 84397), *UAS-Kir2.1::EGFP* (BDSC# 6596), *UAS-Stinger*, *LexAop-tdTomato.nls* (BDSC# 66680), *UAS-mCD8::mCherry*, *UAS-CRTC::GFP* (BDSC# 99657), *UAS-ReaChR* (BDSC# 53741), *UAS-TrpA1* (BDSC# 26263), *LexAop-ReaChR* (BDSC# 53746), *LexAop-rCD2::RFP-p10.UAS-mCD8::GFP-p10* (BDSC# 67093). *w*¹¹¹⁸ was from Y. Rao (Peking University); *HugPC-GAL4* (Hückesfeld et al., 2016 [DOI](#)) was from M. Pankratz (University of Bonn); *LexAop-TrpA1* (Burke et al., 2012 [DOI](#)) was from S. Waddel (University of Oxford). *HugPC-LexA* was generated in this study (see the following section, “Generation of transgenic lines”). Detailed lists of fly genotypes can be found in the key resource table and Source Data files.

General fly maintenance

Flies were maintained on conventional fly food under a 9AM:9PM light/ dark cycle at 25°C and 60 +/- 5 % humidity.

Egg collection

Agar substrates for egg collection were prepared by microwaving a mixture of 2.0 g sucrose (Fujifilm Wako Pure Chemicals, cat# 196-00015), 3.0 g agar (BD, cat#214010), 2.5 mL apple juice (Dole), and Milli-Q water up to 100 mL. Melted agar was poured into Petri dishes with a diameter of 4 cm (approximately 3 mL per dish). Solidified agar was scratched with tweezers to make the surface rough. Agar plates were stored inside a sealed container at 4°C. Upon egg collection, yeast paste (dried yeast (Nippon Beet Sugar Co., Ltd.) kneaded with Milli-Q water at 40% (w/w)) was placed in the center of the agar plates to further promote oviposition.

Larval staging

First-instar larvae were collected two days after egg collection. Among them, second-instar larvae that molted within 2 h were used for video recording. The distinction between first- and second-instar larvae was based on morphological characteristics: while anterior spiracles were not evident in the first-instar, fist-shaped spiracles were observed in the second-instar.

Larval sleep analysis

Twenty-four-well silicone chambers for sleep analysis, described in **Figure 1—figure supplement 1B** [\[1\]](#), were made by pouring dimethylpolysiloxane (PDMS) into molds based on a previous study (Churgin et al., 2019 [\[2\]](#); Szuperak et al., 2018 [\[3\]](#)). As a substrate, 120 µL of a mixture of 3% agar and 2% sucrose was poured into each well. After the gel solidified, 10 µL of 60 mg/mL yeast suspension was applied to the surface of the gel in each well. One larva was transferred into each well, and a glass plate (Azone, cat# 1-4540-01, 5 mm × 200 mm × 200 mm) was placed on top of the chamber to prevent the larvae from escaping. The chamber was placed inside a dark box equipped with a ring-type infrared LED (CCS, cat# LDR2-132IR940-LA) and a camera (The Imaging Source, cat# DMK27BUP031), and recording was started immediately without acclimation. Images were captured through the IC Capture software (The Imaging Source, version 2.5) placed above the chamber under the following conditions: MJPEG Compressor for codec, a frame rate of 0.87 fps, and an image size of 2592 × 1944. For each well and frame, the image of larva was blurred and binarized to automatically locate the larva. Consecutive frames were then compared so that changes in values were summed to estimate the larval motion. This motion estimate was then used to classify every consecutive pair of frames as representing “active” or “inactive.” We found that inactivity across 12 consecutive frames meets the sleep criteria, as shown in **Figure 1—figure supplement 2** [\[4\]](#).

Sleep deprivation

Blue light irradiation was performed using a 455-nm LED (Thorlabs, cat# M455L4) with a collimating adapter (Thorlabs, cat# SM1U25-A). The irradiation intensity was set to the maximum scale using a T-cube LED driver (Thorlabs, cat# LEDD1B). Temporal control of the LED illumination was achieved using an Arduino UNO microcontroller board (Arduino, RRID:SCR_017284). For sleep deprivation experiments, the first 1 h was recorded without LED illumination, followed by another 1 h consisting of repeated cycles of “LED ON for 90 s and OFF for 30 s.”

Immunohistochemistry

The following steps were carried out in succession: dissection in phosphate-buffered saline (PBS); fixation in 4% paraformaldehyde/PBS at room temperature for either 90 min in CRTC::GFP experiments or 30 min in the others; washing with 0.3% Triton X-100/PBS (hereafter referred to as

PBST) for 10 min $\times$ 3; blocking in 5% normal goat serum (NGS)/PBST at room temperature for 30 min; primary antibody treatment at 4°C overnight; washing with PBST for 10 min $\times$ 3; secondary antibody treatment at 4°C overnight; washing with PBST for 10 min $\times$ 3; and mounting in VECTASHIELD medium (Vector Laboratories, cat# H-1000, RRID:AB_2336789). The above steps were performed either in Terasaki 96-well plates (stem Corp., cat# P96R37N) for adults or in 48-well culture plates (Corning, cat# 3548) for larvae. The following primary antibodies were diluted in 5% NGS/PBST: chicken polyclonal anti-GFP (1:1000, Aves Labs, cat# GFP-1020, RRID:AB_10000240), mouse monoclonal anti-mCherry (1:1000, Takara Bio, cat# 632543, RRID:AB_2307319), rabbit polyclonal anti-Dilp3 (1:250, (Veenstra et al., 2008 [DOI](#))), rabbit polyclonal anti-Dilp2 (1:2000, (Okamoto et al., 2012 [DOI](#))), mouse monoclonal anti-BRP (1:100, Developmental Studies Hybridoma Bank, clone nc82; RRID:AB_2314866), and rat monoclonal anti-histone H3 phospho S28 (1:1000, Abcam, cat# ab10543; RRID:AB_2295065). The following secondary antibodies were diluted in 5% NGS/PBST: goat anti-chicken Alexa Fluor 488 (1:200, Thermo Fisher Scientific, cat# A-11039, RRID:AB_2534096), goat anti-mouse Alexa Fluor 555 (1:200, Thermo Fisher Scientific, cat# A-21424, RRID:AB_141780), goat anti-rabbit Alexa Fluor 633 (1:200, Thermo Fisher Scientific, cat# A-21071, RRID:AB_2535732), and goat anti-rat Alexa Fluor 633 (1:200, Thermo Fisher Scientific, cat# A-21094; RRID:AB_2535749). Fluorescence images were obtained using a confocal microscope (Leica, TCS SP8) with 20 $\times$ (Leica, HC PL APO CS2 20 $\times$ /0.75 IMM) and 63 $\times$ (Leica, HC PL APO CS2 63 $\times$ /1.40 OIL) objectives. Images were captured at 512 $\times$ 512 pixel resolution in 1 μ m-steps along the z-axis. Z-stack images of the maximum projections were generated using the ImageJ v1.53t software ((Schneider et al., 2012 [DOI](#)); RRID:SCR_003070).

Generation of transgenic lines

HugPC-LexA lines were generated by Φ C31 integrase-mediated transgenesis, as previously described (Groth et al., 2004 [DOI](#)). The 544-bp enhancer region of *Hug*, identical region to that used for *HugPC-GAL4* (Hückesfeld et al., 2016 [DOI](#)), was PCR-amplified from fly genomic DNA. The gel-purified amplicon was subjected to another round of PCR to add hangout sequences. The backbone suicide-vector pBPnlsLexA::p65Uw (RRID:Addgene_26230) was initially amplified by transforming the *ccdB*-resistant *E. coli* strain, One Shot™ *ccdB* Survival™ 2 T1^R competent cells (Thermo Fisher Scientific, cat# A10460). The PCR product and the vector were digested by Aat II (NEB, cat# R0117S) and Fse I (NEB, cat# R0588S), followed by ligation using In-Fusion HD Cloning Kit w/Cloning Enhancer (Takara Bio, cat# 639635). The integrity of the resulting plasmid was confirmed by DNA sequencing. Plasmid insertion was targeted to four attP sites (attP2, attP40, VK00005, and su(Hw)attP5) individually through Φ C31 integrase-mediated transgenesis by BestGene Inc. (Chino Hills, CA).

Transformants were selected based on the eye color marker. Among the generated strains harboring *HugPC-LexA* in each insertion site, labeling patterns in the larval brain were checked by driving fluorescent marker expression under the control of *LexAop* sequence (**Figure 3—figure supplement 2** [DOI](#)). As the *HugPC-LexA* inserted in attP2 closely resembled the labeling pattern of the original *HugPC-GAL4*, we chose this driver in further experiments.

The following primers were used for cloning:

Forward primer for the *HugPC* enhancer; 1st round PCR against genomic DNA:

5'-AAGGGTTTGGTTTAATTTATTTATGTCATA-3'

Reverse primer for the *HugPC* enhancer; 1st round PCR against the genomic DNA:

5'-GGACAACTGATGCCAGCAGC-3'

Forward primer for the *HugPC* enhancer with hangout sequences (underlined); 2nd round PCR against the 1st round amplicon:

5'-gaaaagtgccacctgacgtAAGGGTTTGGTTTAATTTATTTATGTCATA-3'

Reverse primer for the *HugPC* enhancer with hangout sequences (underlined); 2nd round PCR against the 1st round amplicon:

5'-cccgggcgagctcggccggGGACAACTGATGCCAGCAGC-3'

Measurement of neuronal activity and Dilp3 accumulation in IPCs

The GFP-tagged CRTC (CRTC::GFP) reporter was applied to quantify the activities of larval IPCs, essentially according to the previous report in adult flies (Bonheur et al., 2023 [DOI](#)). Parental *Dilp3-GAL4* and the reporter line harboring *UAS-CRTC::GFP* and *UAS-mCD8::mCherry* (BDSC# 99657) were crossed, and eggs were collected within an 8–10 h time window. Offspring larvae were reared for 4 days after egg laying (AEL) at 25°C until the early third instar. Collected larvae were washed with water and starved on water-soaked paper for 12–14 h. For *in vivo* thermogenetic activation, larva-containing Petri dishes were warmed at 30°C for 60 min. For *ex vivo* thermogenetic experiments, larvae were dissected in calcium- and sugar-free HL3.1 buffer (Nakamizo-Dojo et al., 2023 [DOI](#)) and each brain was incubated in the same buffer at 30°C for 5 min.

For *ex vivo* glucose or peptide application assays, larvae were first dissected in the same buffer as above at room temperature, and brain samples were then treated with either 20 mM D-glucose or 1 mM of chemically synthesized peptides dissolved in HL3.1 buffer for 5 min. After incubation, samples were subjected to immunohistochemical analyses to visualize GFP, mCherry, and anti-Dilp3 signals (see the section “Immunohistochemistry”).

Image processing was performed using ImageJ v1.53t (Schneider et al., 2012 [DOI](#)); RRID:SCR_003070). Nuclear ROIs and cell body outlines were determined based on the mCD8::mCherry fluorescence insensitive to neuronal activity, and the mean intensities of CRTC::GFP signals within the nucleus (F_{nuc}) and the cytoplasm ($F_{\text{cyto}} = F_{\text{whole cell}} - F_{\text{nuc}}$) were quantified. NLI was calculated using the following formula: $(F_{\text{nuc}} - F_{\text{cyto}})/(F_{\text{nuc}} + F_{\text{cyto}})$. To assess Dilp3 accumulation in IPCs, the anti-Dilp3 signal intensity was measured within the cytosol of each cell, excluding the nucleus. Data from 4–14 Dilp3-positive IPCs per brain, collected from 3–7 larvae, were pooled for each genotype/treatment group.

Chemical synthesis of Hug peptides

Peptides with the following amino acid sequences were chemically synthesized using the Fmoc solid phase method, and their purity was assessed by HPLC to be over 95%. All procedures were carried out by Eurofins Genomics K.K. (Tokyo, Japan). The amino acid sequences of the peptides used were as follows:

- *Drosophila* Hug-γ:
 - Acetyl-QLQSNGEPAYRVTPRL-CONH₂ (17 a.a.)
- *Drosophila* PK-2:
 - Acetyl-SVPFKPRL-CONH₂ (8 a.a.)
- Human NMU:
 - Acetyl-FRVDEEFQSPFASQSRGYFLFRPRN-CONH₂ (25 a.a.)

Sleep quantification in adult flies

The following parts of the *Drosophila* activity monitor (DAM) system were purchased from TriKinetics Inc. (Waltham, MA): the *Drosophila* Activity Monitor (cat# DAM2), the power supply interface unit (cat# PSIU9), monitor tubes with an outside diameter of 5 mm and a length of 65 mm (cat# PGT5x65), and tube caps (cat# CAP5-Black). Newly eclosed adult flies were collected under brief CO₂ anesthesia and kept as groups of 10–15 in standard food vials. Males and females were kept separately to avoid mating. To minimize disturbance by noise and vibration, flies were placed inside a fan-less Peltier-type incubator (Mitsubishi Electric Engineering Co., Ltd., cat# SLC-25) set to 25°C, externally connected with 5-mm diameter LED lights (MY-CRAFT, Ltd., cat# 5W0601) and a timer device (Panasonic Corp., cat# WH3311WP) to generate a 9AM:9PM light/dark cycle. After 3 days of eclosion, each fly was gently aspirated into a Pyrex glass monitor tube, one end of which was stuffed with agar food (2% (w/w) agar and 5% (w/w) sucrose), and the other was plugged with a cotton string. The tubes were set to the monitor device and placed inside the same fan-less incubator. The DAM system was connected to an external PC running DAMSystem3 Data Collection Software (RRID:SCR_021809; available at <https://trikinetics.com/>), which counted beam-crossings by each fly for two successive days. Time windows longer than 5 min without beam-crossing were judged as sleep bouts, following the conventional criteria for adult fly sleep (Paul et al., 2000).

Statistical analysis

Statistical analyses by Kruskal–Wallis one-way ANOVA, Mann–Whitney U-test, or chi-squared test were performed using Prism 9.5.1 (GraphPad, RRID:SCR_002798). Bonferroni correction was applied for multiple comparisons. Asterisks (*) represent p-values as indicated within each figure legend. All data necessary to reproduce the figure panels and statistical analyses are available as Source Data files.

Figure supplements

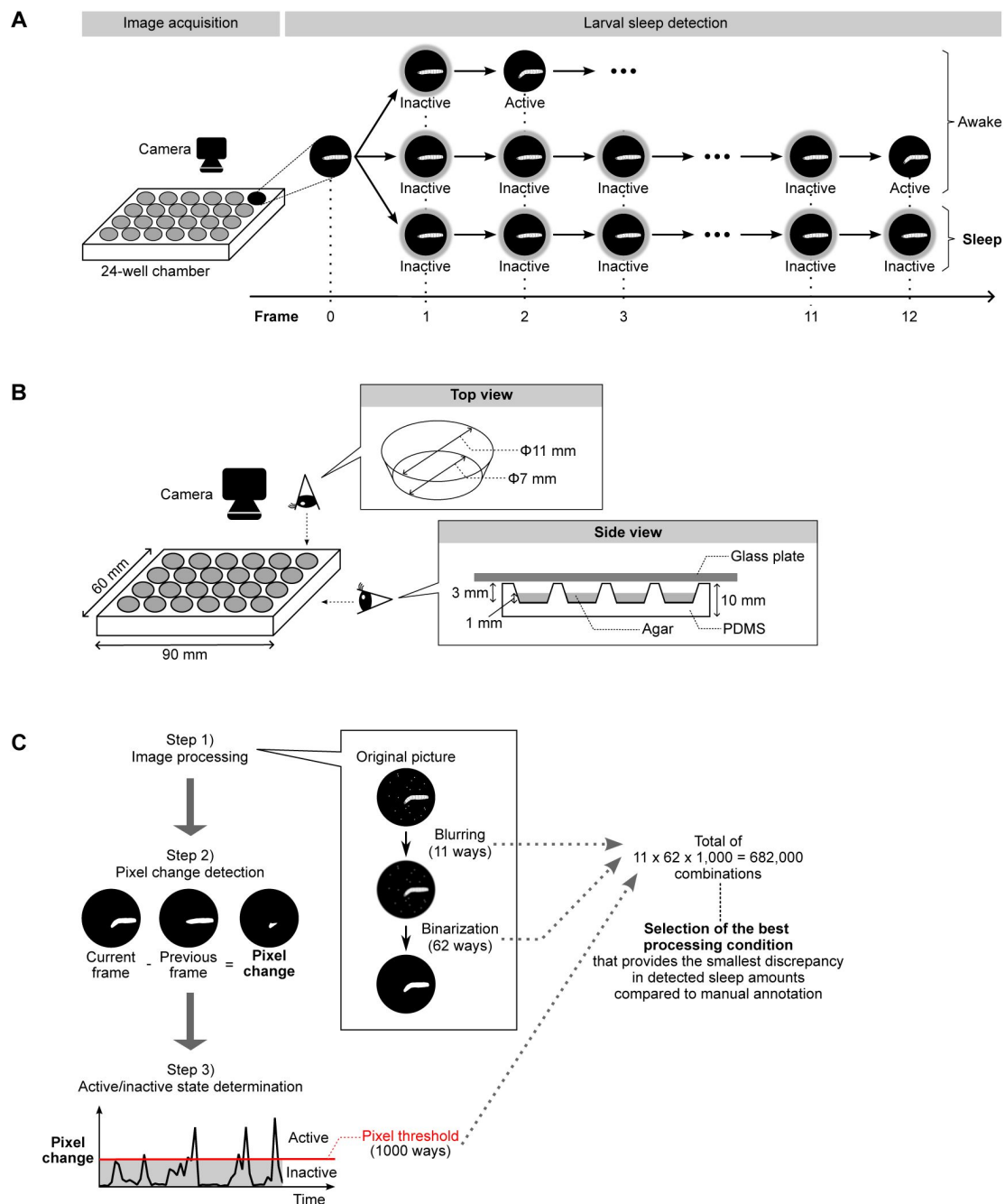


Figure 1—figure supplement 1.

Detailed experimental setup for analyzing larval sleep

(A) Schematic flow of automated larval sleep quantification. A second instar larva was placed in each of 24 wells, and all 24 wells were videotaped from above. For each well and frame, the image of larva was blurred and binarized to automatically locate the larva. Consecutive frames were then compared so that changes in pixel values were summed to estimate larval motion. This motion estimate was then used to classify every consecutive pair of frames as representing “active” or “inactive.” We found that inactivity across 12 consecutive frames meets the sleep criteria, as shown in **Figure 1—figure supplement 2**. (B) Schematic representation of the 24-well chamber used for larval sleep assays. (C) Flow of automated processes of larval motion detection and active/inactive state determination. For each well and frame, the image of larva was blurred and binarized. Consecutive frames were then compared so that changes in pixel values were summed to estimate larval motion. This motion estimate was then used to classify every consecutive pair of frames as representing “active” or “inactive.”

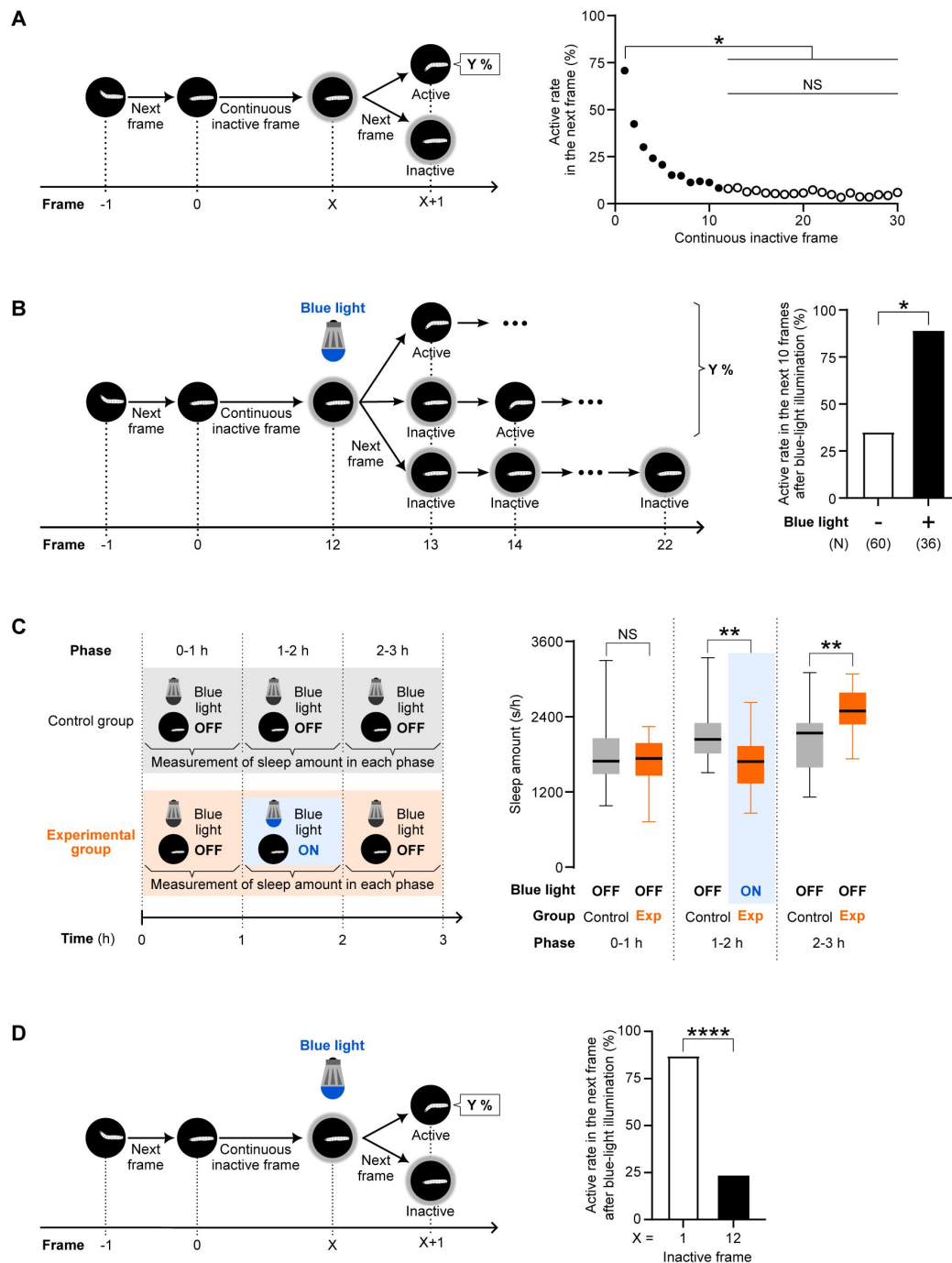


Figure 1—figure supplement 2.

Larval "sleep state" in this study is consistent with behavioral criteria for sleep

(A) Relationship between "inactive" state duration and spontaneous arousal probability. Sustained quiescence for ≥ 12 frames (open circles) significantly reduced arousal probability compared to 1-frame inactivity. * $p < 0.05$, NS: $p \geq 0.05$ (chi-squared test). (B) The percentage of larvae that showed a response within the next 10 frames after high-intensity blue light. The number of events analyzed was 36 and 60 with and without blue light irradiation, respectively. * $p < 0.05$ (Fisher's exact test). (C) Sleep amounts before, during, and after blue light irradiation. The number of larvae tested was 21 and 21 for the control and experimental groups, respectively. ** $p < 0.01$, NS: $p \geq 0.05$ (Mann-Whitney U-test with Bonferroni correction). (D) Percentage of larvae that showed a response in the next 1 or 12 frames after blue light stimulation. Number of events analyzed: 1960 or 111 for 1- or 12-frame continuous inactivity prior to blue light illumination, respectively. **** $p < 0.0001$ (chi-squared test).

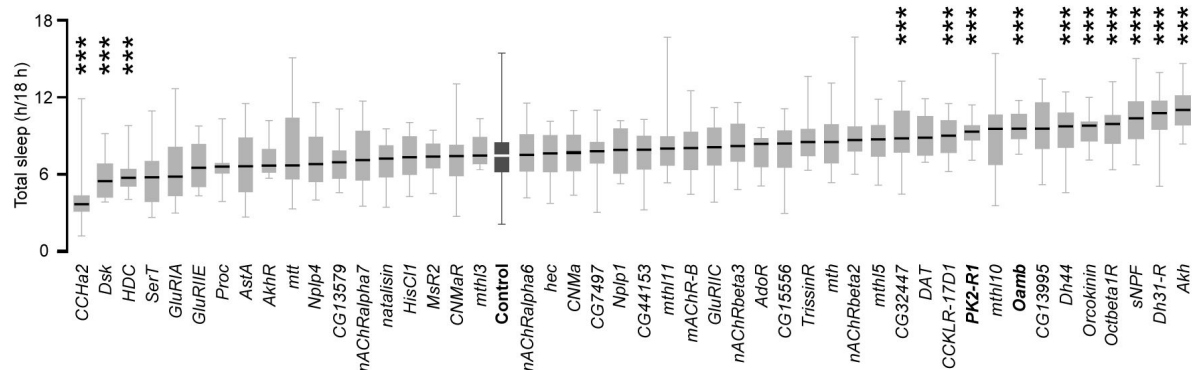


Figure 1—figure supplement 3.

CRISPR-knock-out screen for genes that regulate larval sleep.

Total sleep amounts in CRISPR-knock-out mutants were measured for 18 h from the beginning of the second instar. The data for Control, *PK2-R1*, and *Oamb* mutants are identical to those shown in [Figure 1](#). Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.005$ (Kruskal-Wallis one-way ANOVA and post hoc Mann-Whitney U-test with Bonferroni correction). Number of larvae tested: 475 for the control and 10–52 for each mutant.

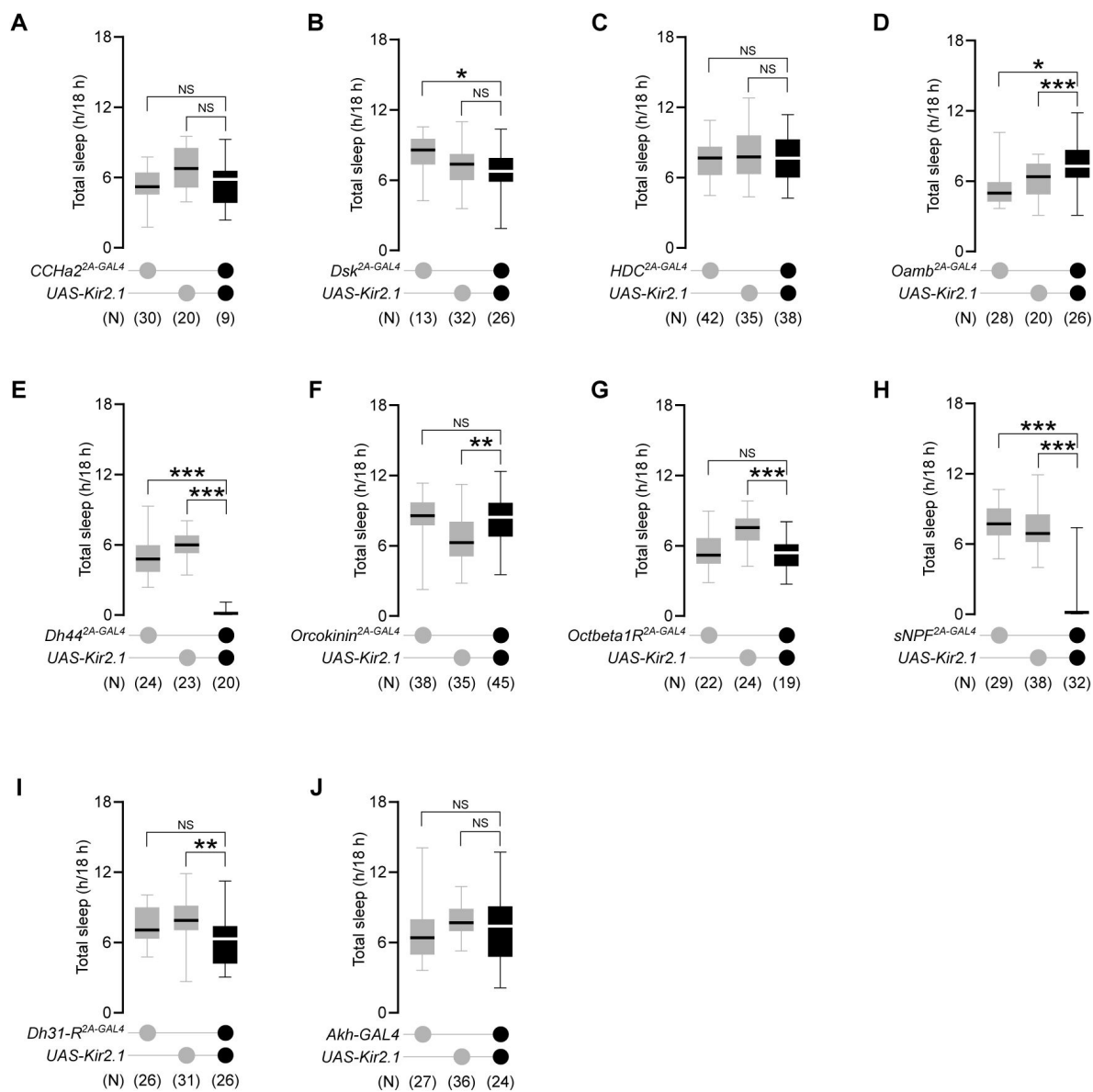


Figure 1—figure supplement 4.

Effects of neuronal manipulation on larval sleep

Sleep amounts in larvae in which distinct neuronal populations were silenced. Kir2.1 expression was driven by each enhancer-GAL4 for “hit” genes identified in the CRISPR-knock-out screen. ‘N’ indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS: ≥ 0.05 (Mann-Whitney U-test with Bonferroni correction).

Figure 2—figure supplement 1.

Locomotion speeds are not consistently affected by genetical manipulations of *PK2-R1*, IPCs, or *Dilps* (A–B) Effects of silencing *PK2-R1* neurons (A) and IPCs (B) on larval locomotion speed. (C) Larval locomotion speed in null mutants of *Dilp3* or *Dilp5*. The average locomotion speed during the wake periods was measured from the data presented in **Figures 1B**, **2C**, and **2D**. In this and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** $p < 0.0001$, * $p < 0.05$, NS: $p \geq 0.05$ (Mann–Whitney U-test with Bonferroni correction).

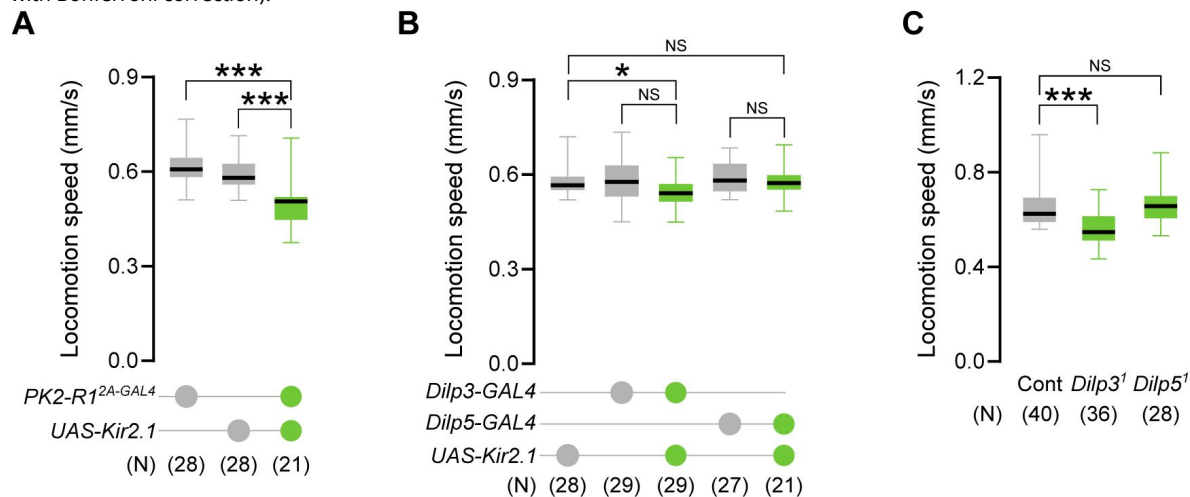
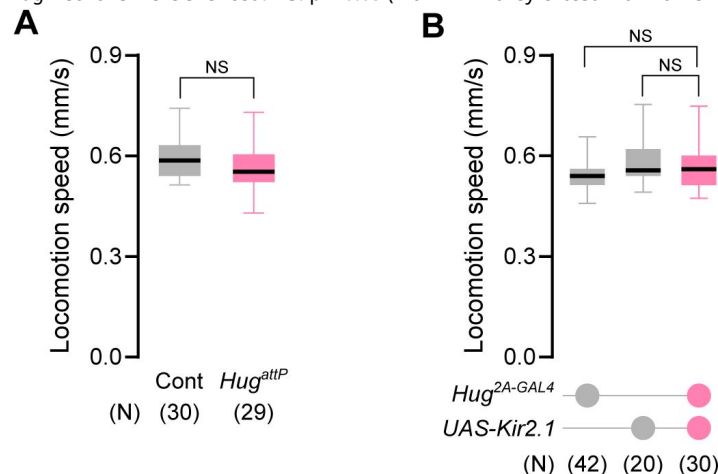


Figure 3—figure supplement 1.

Neither *Hug* knock-out mutation nor *Hug* neuron silencing significantly affects larval locomotion speed (A) Average locomotion speed of *Hug* knock-out mutants during waking periods. The data presented in **Figures 3A** and **3B** were reanalyzed. In this and the following panels, 'N' indicates the number of biologically independent animals used for each group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. NS: $p \geq 0.05$ (Mann–Whitney U-test). (B) Locomotion speed of larvae in which *Hug* neurons were silenced. NS: $p \geq 0.05$ (Mann–Whitney U-test with Bonferroni correction).



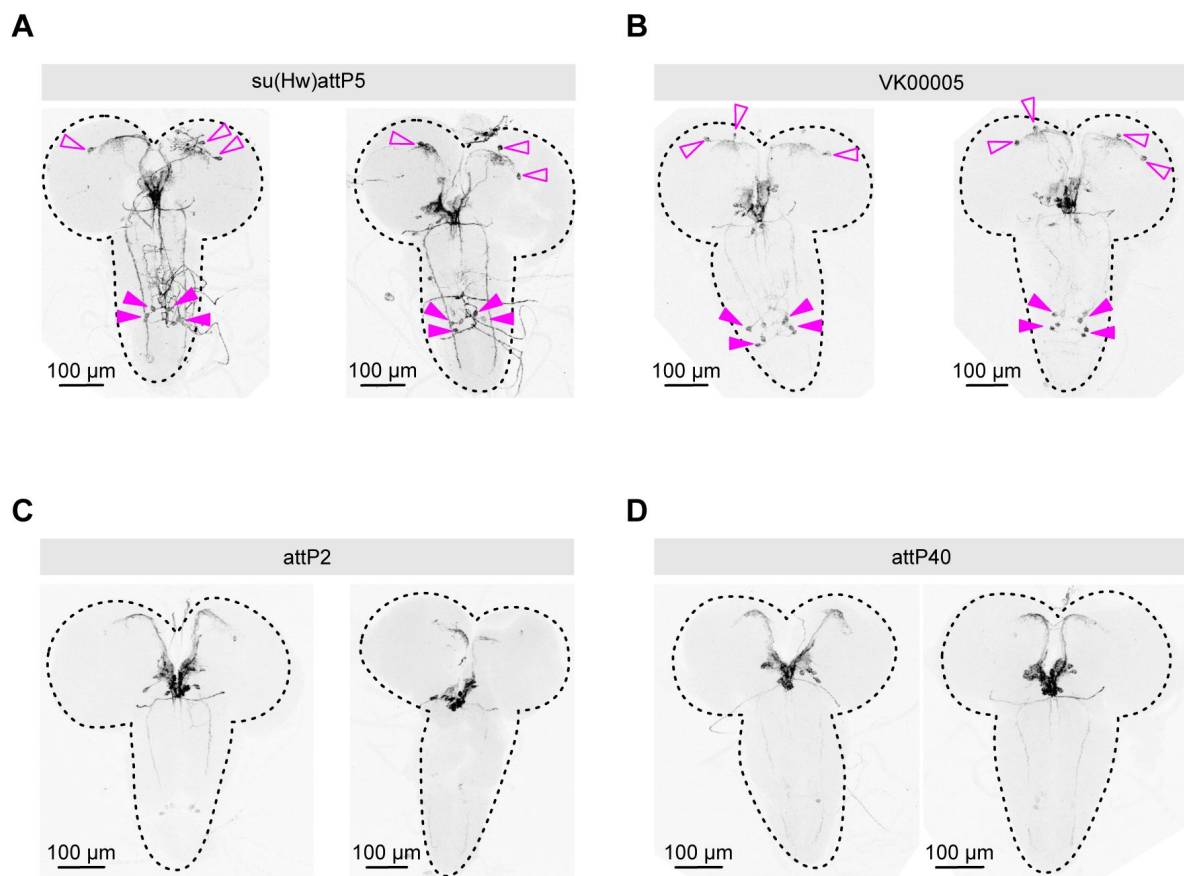


Figure 3—figure supplement 2.

Generation of *HugPC-LexA* transgenic lines

(A-D) Labeling patterns of four *HugPC-LexA* strains in larval CNS visualized by rCD2::RFP. These transgenic lines were generated by injecting the identical plasmid into the following landing sites, respectively: su(Hw)attP5, VK00005, attP2, and attP40. Arrowheads indicate cell bodies of non-HugPC neurons. Filled and unfilled arrowheads indicate cell bodies within the VNC and near the protocerebrum, respectively.

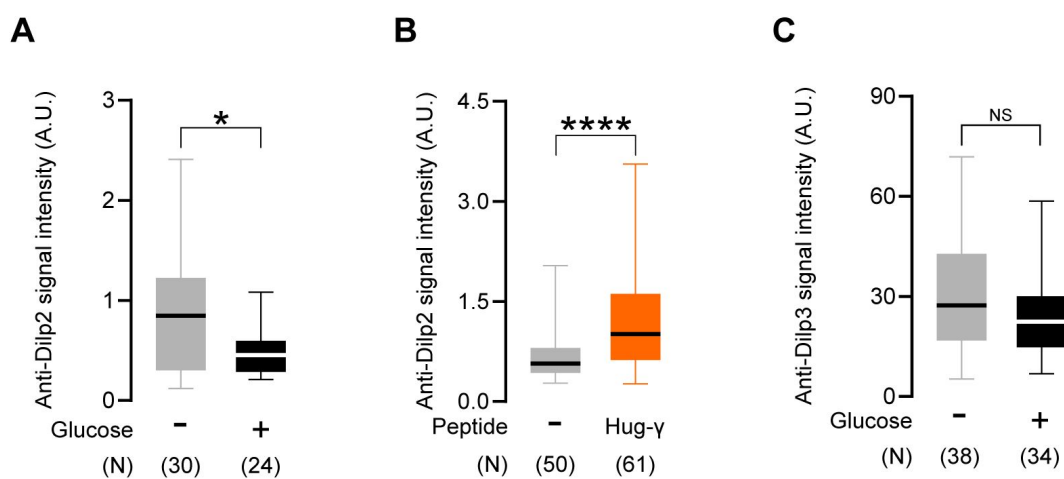


Figure 5—figure supplement 1.

D-glucose but not Hug peptide cause Dilp2 reduction in IPCs

(A–B) Anti-Dilp2 signal intensity measured in IPCs after glucose (A) or Hug-γ application (B). In this and the following panels, ‘N’ indicates the number of cells used for each group. Box plots are generated so that center line indicates median, box limits indicate upper and lower quartiles, and whiskers indicate the minimum-to-maximum range. **** $p < 0.0001$, * $p < 0.05$ (Mann–Whitney U-test). (C) Anti-Dilp3 signal intensity measured within the cytosolic areas of IPCs. NS: ≥ 0.05 (Mann–Whitney U-test).

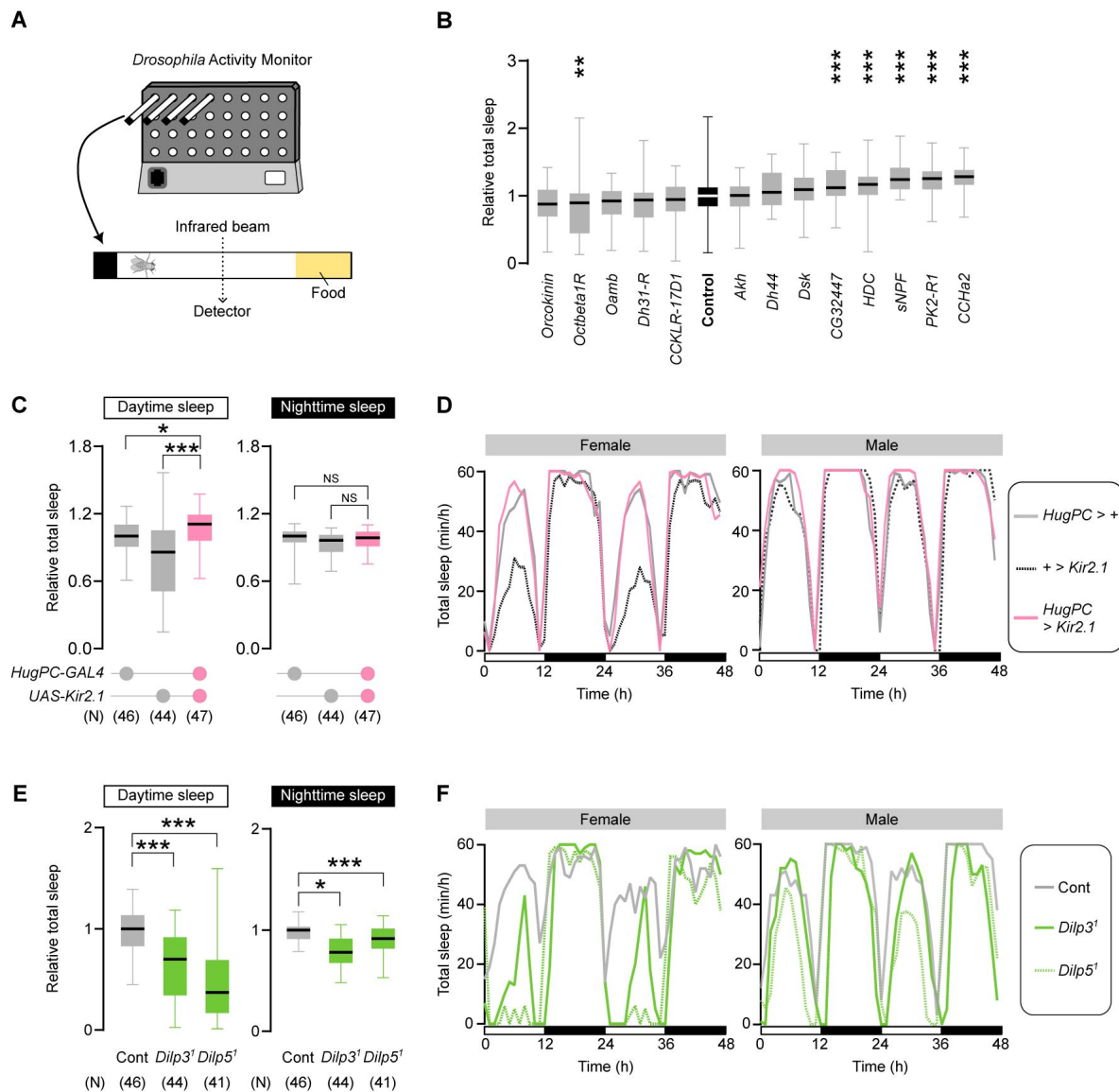


Figure 6—figure supplement 1.

Sleep patterns of adult flies following manipulations of Hugin and insulin pathways

(A) Experimental setup for monitoring adult sleep using the *Drosophila* Activity Monitor (DAM). (B) Sleep duration in adult CRISPR-knock-out mutants. These mutants showed significant changes in the sleep amounts in larvae (**Figure 1—figure supplement 3**). In panels B, C, and E, sleep amounts were calculated as the ratio relative to the median sleep amount of the control group. Box plots are generated so that the center line indicates the median, the box limits indicate the upper and lower quartiles, and the whiskers indicate the minimum-to-maximum range. *** p < 0.001, ** p < 0.01 (Kruskal-Wallis one-way ANOVA and post hoc Mann-Whitney U-test with Bonferroni correction). (C, E) Sleep amounts relative to the median value of the control group. Data presented in **Figures 6B** and **6C** were reanalyzed. *** p < 0.001, * p < 0.05, NS: p ≥ 0.05 (Mann-Whitney U-test with Bonferroni correction). (D, F) Temporal sleep patterns over two consecutive days.

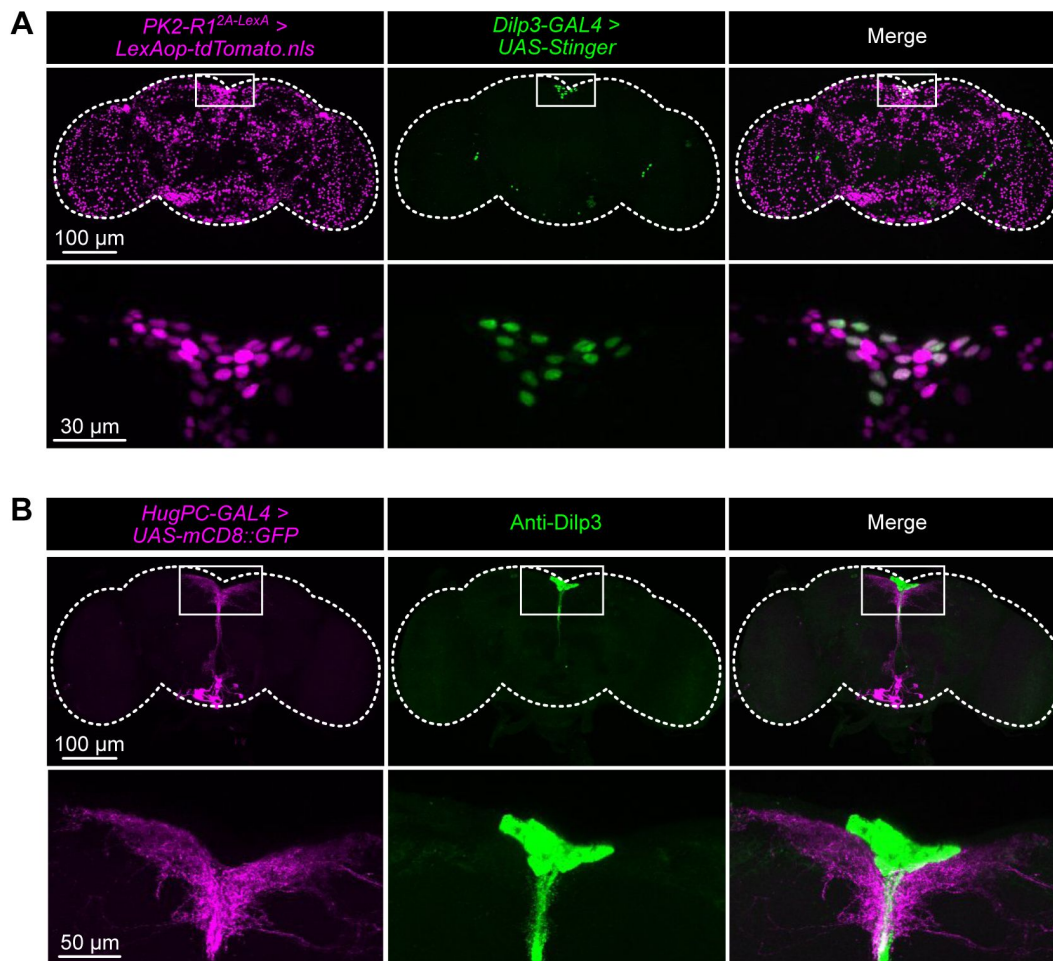


Figure 6—figure supplement 2.

Morphology of *PK2-R1* neurons, HugPC neurons, and IPCs in the adult brain

(A) Dual labeling of *PK2-R1* neurons and IPCs expressing nuclear-localized RFP (stinger; magenta) and GFP (stinger; green), respectively. The top panels show signals in the entire adult brain. The bottom panels show magnified images of the white-squared area in each corresponding top panel, where the cell bodies of IPCs are located. Similar results were obtained across five independent samples. (B) Visualization of HugPC neurons and IPCs labeled by mCD8::GFP (magenta) and the anti-Dilp3 antibody (green), respectively. Similar results were obtained across four independent samples.

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Additional information

Author contributions

C.H., K.I., and M.M. performed experiments and data analyses; M.T. wrote the codes for automated larval detection and sleep quantification; C.H., K.I., and K.E. are responsible for study conception, experimental design, and data interpretation; C.H., K.I., and K.E. wrote the original manuscript, and M.T. edited the paper.

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Reviewer #1 (Public review):

Summary:

The study investigates how neuropeptidergic signaling affects sleep regulation in *Drosophila* larvae. The authors first conduct a screen of CRISPR knock-out lines of genes encoding enzymes or receptors for neuropeptides and monoamines. As a result of this screen, the authors follow up on one hit, the hugin receptor, PK2-R1. They use genetic approaches, including mutants and targeted manipulations of PK2-R1 activity in insulin-producing cells (IPCs) to increase total sleep amounts in 2nd instar larvae. Similarly, *dilp3* and *dilp5* null mutants and genetic silencing of IPCs show increases in sleep. The authors also show that hugin mutants and thermogenetic/optogenetic activation of hugin-expressing neurons caused reductions in sleep. Furthermore, they show through imaging-based approaches that hugin-expressing neurons activate IPCs. A key finding is that wash-on of hugin peptides, Hug-γ and PK-2, in ex vivo brain preparations activates larval IPCs, as assayed by CRTC::GFP imaging. The authors then examine how the PK2-R1, hugin, and IPC manipulations affect adult sleep. Finally, the authors examine how Ca²⁺ responses through CRTC::GFP imaging in adult IPCs are influenced by the wash-on of hugin peptides. The conclusions of this paper are somewhat well supported by data, but some aspects of the experimental approach and sleep analysis need to be clarified and extended.

Strengths:

(1) This paper builds on previously published studies that examine *Drosophila* larval sleep regulation. Through the power of *Drosophila* genetics, this study yields additional insights into what role neuropeptides play in the regulation of *Drosophila* larval sleep.

(2) This study utilizes several diverse approaches to examine larval and adult sleep regulation, neural activity, and circuit connections. The impressive array of distinct analyses provides new understanding into how *Drosophila* sleep-wake circuitry is regulated across the lifespan.

(3) The imaging approaches used to examine IPC activation upon hugin manipulation (either thermogenetic activation or wash-on of peptides) demonstrate a powerful approach for examining how changes in neuropeptidergic signaling affect downstream neurons. These experiments involve precise manipulations as the authors use both *in vivo* and *ex vivo* conditions to observe an effect on IPC activity.

Weaknesses:

Although the paper does have some strengths in principle, these strengths are not fully supported by the experimental approaches used by the authors. In particular:

(1) The authors show total sleep amount over an 18-hour period for all the measures of 2nd instar larval sleep throughout the paper. However, published studies have shown that sleep changes over the course of 2nd instar development, so more precise time windows are necessary for the analyses in this study.

(2) Previously published reports of sleep metrics in both *Drosophila* larvae and adults include the average number of sleep episodes (bout number) and the average length of sleep episodes (bout length). Neither of these metrics is included in the paper for either the larval sleep or adult sleep data. Not including these metrics makes it difficult for readers to compare the findings in this study to previously published papers in the established *Drosophila* sleep literature.

(3) Because *Drosophila* adult & larval sleep is based on locomotion, the authors need to show the activity values for the experiments supporting their key conclusions. They do show travel distances in Figure 2 - Figure Supplement 1, however, it is not clear how these distances were calculated or how the distances relate to the overall activity of individual larvae during sleep experiments. It is also concerning that inactivation of the PK2-R1-expressing neurons causes a reduction in locomotion speed. This could partially explain the increase in sleep that they observe.

(4) The authors rely on homozygous mutant larvae and adult flies to support many of their conclusions. They also rely on Gal4 lines with fairly broad expression in the *Drosophila* brain to support their conclusions. Adding more precise tissue-specific manipulations, including thermogenetic activation and inhibition of smaller populations of neurons in the study would be needed to increase confidence in the presented results. Similarly, demonstrating that larval development and feeding are not affected by the broad manipulations would strengthen the conclusions.

(5) Many of the experiments presented in this study would benefit from genetic and temperature controls. These controls would increase confidence in the presented results.

(6) The authors claim that their findings in larvae uncover the circuit basis for larval sleep regulation. However, there is very little comparison to published studies demonstrating that neuropeptides like Dh44 regulate larval sleep. Because hugin-expressing neurons have been shown to be downstream of Dh44 neurons, the authors need to include this as part of their discussion. The authors also do not explain why other neuropeptides in the initial screen are not pursued in the study. Given the effect that these manipulations have on larval sleep in their initial screen, it seems likely that other neuropeptidergic circuits regulate larval sleep.

<https://doi.org/10.7554/eLife.105710.1.sa2>

Reviewer #2 (Public review):

Summary:

This study examines larval sleep patterns and compares them to sleep regulation in adult flies. The authors demonstrate hallmark sleep characteristics in larvae, including sleep rebound and increased arousal thresholds. Through genetic and behavioral analyses, they identify PK2-R1 as a key receptor involved in sleep modulation, likely via the HuginPC-IPC signaling pathway. Loss of PK2-R1 results in increased sleep, which aligns with previous findings in hugin knockout mutants. While the study presents significant contributions to the field, further investigation is needed to address discrepancies with earlier research and strengthen mechanistic claims.

Strengths:

- (1) The study explores a relatively understudied aspect of sleep regulation, focusing on larval development.
- (2) The use of an automated behavioral measurement system ensures precise quantification of sleep patterns.
- (3) The findings provide strong genetic and behavioral evidence supporting the role of the HuginPC-IPC pathway in sleep regulation.
- (4) The study has broader implications for understanding the evolution and functional divergence of sleep circuits.

Weaknesses:

- (1) The manuscript does not sufficiently discuss previous studies, particularly concerning hugin mutants and their metabolic effects.
- (2) The specificity of IPC secretion mechanisms is unclear, particularly regarding potential indirect effects on Dilp2.
- (3) Alternative circuits, such as the HuginPC-DH44 pathway, require further consideration.
- (4) Functional connectivity between HuginPC neurons and IPCs is not directly validated.
- (5) Developmental differences in sleep regulatory mechanisms are not thoroughly examined.

<https://doi.org/10.7554/eLife.105710.1.sa1>

Reviewer #3 (Public review):

Summary:

Sleep affects cognition and metabolism, evolving throughout development. In mammals, infants have fast sleep-wake cycles that stabilize in adults via circadian regulation. In this study, the author performed a genetic screen for neurotransmitters/peptides regulating sleep and identified the neuropeptide Hugin and its receptor PK2-R1 as essential components for sleep in *Drosophila* larvae. They showed that IPCs express Pk2-R1 and silencing IPCs resulted in a significant increase in the sleep amount, which was consistent with the effect they observed in PK2-R1 knock-out mutants. They also showed that Hugin peptides, secreted by a subset of Hugin neurons (Hug-PC), activate IPCs through the PK2-R1 receptor. This activation prompts IPCs to release insulin-like peptides (Dilps), which are implicated in the modulation of sleep. They showed that Hugin peptides induce a PK2-R1 dependent calcium (Ca^{2+})

increase in IPCs, which they linked to the release of Dilp3, showing a connection between Hugin signaling to IPCs, Dilp3 release, and sleep regulation. Additionally, the activation of Hug-PC neurons reduced sleep amounts, while silencing them had the opposite effect. In contrast to the larval stage, the Hugin/PK2-R1 axis was not critical for sleep regulation in *Drosophila* adults, suggesting that this neuropeptidergic circuitry has divergent roles in sleep regulation across different stages of development.

Strengths:

This study used an updated system for sleep quantification in *Drosophila* larvae, and this method allowed precise measurement of larval sleep patterns which is essential for the understanding of sleep regulation.

The authors performed unbiased genetics screening and successfully identified novel regulators for larval sleep, Hugin and its receptor PK2-R1, making a substantial contribution to the understanding of neuropeptidergic control of sleep regulation.

They clearly demonstrated the mechanism by which Hugin-expressing neurons influence sleep through the activation of IPCs via PK2-R1 with Ca²⁺ responses and can modulate sleep.

Based on the demonstrated activation of PK2-R1 by the human Hugin orthologue Neuromedin U, research on human sleep disorders may benefit from the discoveries from *Drosophila* since sleep-regulating mechanisms are conserved across species.

Weaknesses:

The study primarily focused on sleep regulation in *Drosophila* larvae, showing that the Hugin/PK2-R1 axis is critical for larval sleep but not necessary for adult sleep. The effects of the Hugin axis in the adult are, however, incompletely explained and somewhat inconsistent. PK2-R1 knockout adults also display increased sleep, as does HugPC silencing, at least for daytime sleep. The difference lies in Dilp3/5 mutant animals showing decreased sleep and IPCs seemingly responding with reduced Dilp3 release to PK-2 treatment (Figure 6). It seems difficult to reconcile the author's conclusions regarding this point without additional data. It could be argued that PK2-R1 still regulates adult sleep, but not via Hugin and IPCs/Dilps.

Another issue might be that the authors show relative sleep levels for adults using Trikinetics monitoring. From the methods, it is not clear if the authors backcrossed their line to an isogenic wild-type background to normalize for line-specific effects on sleep. Thus, it is likely that each line has differences in total sleep time due to background effects, e.g., their Kir2.1 control line showed reduced sleep relative to the compared genotypes. This might limit the conclusions on the role of Hugin/PK2-R1 on adult sleep.

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